Journal of Experimental Biology

Size-specific strategies of sympatric cetaceans to reduce heat loss

--Manuscript Draft--

Manuscript Number:	
Article Type:	Research Article
Full Title:	Size-specific strategies of sympatric cetaceans to reduce heat loss
Abstract:	Maintaining a stable core body temperature is essential for endotherms. Cetaceans live in a highly thermally conductive medium, requiring special adaptations to reduce heat loss and maintain homeothermy. We employed a combination of aerial photogrammetry and existing data sources to estimate heat loss rates in five sympatric cetaceans of varying sizes, inhabiting the sub-arctic waters (~3.7 °C) of NE-Iceland: harbour porpoises (Phocoena phocoena, 0.97 - 1.60 m, n = 50), white-beaked dolphins (Lagenorhynchus albirostris, 1.10 - 2.86 m, n = 294), minke whales (Balaenoptera acutorostrata, 4.36 - 8.64 m, n = 30), humpback whales (Megaptera novaeangliae, 6.01 - 14.2 m, n = 282) and blue whales (Balaenoptera musculus, 13.17 - 24.23 m, n = 29). Further, we investigated the effect of body size (length), body shape (surface-area-to-volume ratio, SVR), body temperature, and blubber thermal conductivity and thickness on heat loss for all species. Smaller species had higher volume-specific heat loss compared to larger species due to their higher SVRs, a fundamental consequence of scaling. Apart from body size, blubber thickness had the largest effect on heat loss, followed by thermal conductivity. Smaller cetaceans seem to rely primarily on physiological and morphological adaptations to reduce heat loss, such as increased blubber thickness and lower thermal conductivity, whereas larger species offset heat loss by having larger bodies and lower SVRs. Our findings are critical for assessing the vulnerability of these species to climate change, as they provide valuable insights into their thermal biology and its implications for habitat use and prey requirements.
Corresponding Author:	Maria Glarou University of Iceland: Haskoli Islands Húsavík, ICELAND
Other Authors:	Marianne Helene Rasmussen
	Alyssa Poldner
	Silva Naomi Sophie Ruppert
	Dido Sotiropoulou
	Zaynab Sadozai
	Małgorzata Jarzynowska
	Maria Refsgaard Iversen
	Guðjón Már Sigurðsson
	Sverrir Daníel Halldórsson
	Valerie Chosson
	Phillip J. Clapham
	Yulia Ivashchenko
	Dmitry Tormosov
	Fredrik Christiansen
Keywords:	allometry; cetaceans; ecophysiology; heat loss; thermoregulation; UAV photogrammetry
Manuscript Classifications:	Mammal; Ecological and evolutionary physiology
Suggested Reviewers:	Magnus Wahlberg Associate Professor, University of Southern Denmark: Syddansk Universitet magnus@biology.sdu.dk

Powered by Editorial Manager® and ProduXion Manager® from Aries Systems Corporation

	Christina Lockyer Research Director, Age Dynamics christina_lockyer@hotmail.com Expert in large whale bioenergetics
	Michael Moore Senior Scientist, Woods Hole Oceanographic Institution mmoore@whoi.edu
Opposed Reviewers:	
Additional Information:	
Question	**Response**
Companion Paper Is your paper a companion paper (part of a group of papers being submitted)?	No
Editor Suggestions You may request that your submission is assigned to a specific editor. Please suggest no more than three editor names in order of preference. Although you may suggest an editor for your submission, the journal will make the final assignment. If you do not request an editor, your submission will be assigned to the most appropriate editor as determined by the editorial staff.	N/A
Special Issue	No, this article is not part of a special themed issue
Open Access Please check your funder requirements carefully and select the appropriate Open Access publication route below. A quote for your Open Access publication charges will be available to view on the final submission page, revealing any institutional discounts and providing you with another chance to select Open Access. See **Instructions** for further information.	**Green Open Access**
Word Count	6038
Number of Figures	6
Number of Tables	3

Powered by Editorial Manager® and ProduXion Manager® from Aries Systems Corporation

UNIVERSITY OF ICELAND

Saemundargata 2, 102 Reykjavik, Iceland
Tel. +354 525 4000 · hi@hi.is · hi.is

July 31th, 2024

Dear Editor of the *Journal of Experimental Biology*,

We would be very grateful if you would consider our manuscript entitled "*Size-specific strategies of sympatric cetaceans to reduce heat loss*" for publication as a research article in the *Journal of Experimental Biology*.

Maintaining a stable core body temperature is essential for the survival of endotherms. Since cetaceans live in a highly thermally conductive medium, they require special adaptations to reduce heat loss and maintain homeothermy. In this study, we employed a combination of drone aerial photogrammetry and existing data sources to estimate and compare the heat loss rates in five sympatric cetaceans of varying sizes (ranging from harbour porpoises to blue whales), all inhabiting the cold, sub-arctic waters of NE-Iceland. We further investigated the effects of body size (length), body shape (surface-area-to-volume ratio, SVR), body temperature, and blubber thermal conductivity and thickness on heat loss for all target species. Overall, we found that smaller cetaceans rely primarily on physiological and morphological adaptations, such as increased blubber thickness and lower thermal conductivity, to reduce heat loss. In contrast, larger species offset heat loss through their larger bodies and lower SVRs. Our findings are critical for assessing the vulnerability of these species to climate change, providing valuable insights into their thermal biology and its implications for habitat use and prey requirements. Despite focusing on five cetacean species in this study, our novel drone aerial photogrammetry method for estimating surface area and heat loss has the potential to be applied to other aquatic mammals. Hence, we believe that both the methodological and ecological aspects of this study will interest the broader readership of the *Journal of Experimental Biology*.

Our manuscript represents original work, is not under consideration for publication elsewhere and has been read and agreed upon by all authors. There are no conflicts of interest to disclose. All sources of funding are acknowledged in the manuscript. We hope that you will find our manuscript suitable for publication in *Journal of Experimental Biology* and thank you for your time, consideration and effort.

Yours sincerely on behalf of the authors,



Maria Glarou
PhD fellow
Húsavík Research Centre, University of Iceland
Hafnarstétt 3
640 Húsavík, Iceland
maria_glarou@outlook.com
+30 6985689823

CONFIDENTIAL

Size-specific strategies of sympatric cetaceans to reduce heat loss

Running title: Heat loss strategies in sympatric cetaceans

Maria Glarou[1,*], Marianne Helene Rasmussen[1], Alyssa Poldner[2], Silva Naomi Sophie Ruppert[3], Dido Sotiropoulou[4], Zaynab Sadozai[5], Małgorzata Jarzynowska[6], Maria Refsgaard Iversen[1], Guðjón Már Sigurðsson[7], Sverrir Daníel Halldórsson[7], Valerie Chosson[7], Phillip J. Clapham[8], Yulia Ivashchenko[8], Dmitry Tormosov[9], Fredrik Christiansen[10]

[1] Húsavík Research Centre, University of Iceland, 640 Húsavík, Iceland

[2] Department of Wildlife Management, Van Hall Larenstein University of Applied Sciences, 8934 CJ, Leeuwarden, The Netherlands

[3] Institute for Biology and Environmental Sciences, Carl von Ossietzky University Oldenburg, 26129 Oldenburg, Germany

[4] Behavioural Ecology Group, Section for Ecology and Evolution, Department of Biology, University of Copenhagen, 2100 Copenhagen, Denmark

[5] Department of Biology, University of Cologne, 50674 Cologne, Germany

[6] Faculty of Oceanography and Geography, University of Gdańsk, 81-378 Gdynia, Poland

[7] Pelagic Division, Marine and Freshwater Research Institute, 220 Hafnarfjordur, Iceland

[8] Seastar Scientific, Vashon, WA 98070, USA

[9] Seastar Scientific Russia, Kaliningrad, Karl Marx St 76, 236022 Russia

[10] Marine Mammal Research, Department of Ecoscience, Aarhus University, 4000 Roskilde, Denmark

*Corresponding Author

E-mail: maria_glarou@outlook.com

Address: Húsavík Research Centre, University of Iceland, 640 Húsavík, Iceland

Tel: +45 50 17 40 14

CONFIDENTIAL

Abstract

Maintaining a stable core body temperature is essential for endotherms. Cetaceans live in a highly thermally conductive medium, requiring special adaptations to reduce heat loss and maintain homeothermy. We employed a combination of aerial photogrammetry and existing data sources to estimate heat loss rates in five sympatric cetaceans of varying sizes, inhabiting the sub-arctic waters (~3.7 °C) of NE-Iceland: harbour porpoises (*Phocoena phocoena*, 0.97 - 1.60 m, n = 50), white-beaked dolphins (*Lagenorhynchus albirostris*, 1.10 - 2.86 m, n = 294), minke whales (*Balaenoptera acutorostrata*, 4.36 - 8.64 m, n = 30), humpback whales (*Megaptera novaeangliae*, 6.01 - 14.2 m, n = 282) and blue whales (*Balaenoptera musculus*, 13.17 - 24.23 m, n = 29). Further, we investigated the effect of body size (length), body shape (surface-area-to-volume ratio, SVR), body temperature, and blubber thermal conductivity and thickness on heat loss for all species. Smaller species had higher volume-specific heat loss compared to larger species due to their higher SVRs, a fundamental consequence of scaling. Apart from body size, blubber thickness had the largest effect on heat loss, followed by thermal conductivity. Smaller cetaceans seem to rely primarily on physiological and morphological adaptations to reduce heat loss, such as increased blubber thickness and lower thermal conductivity, whereas larger species offset heat loss by having larger bodies and lower SVRs. Our findings are critical for assessing the vulnerability of these species to climate change, as they provide valuable insights into their thermal biology and its implications for habitat use and prey requirements.

Keywords: allometry; cetaceans; ecophysiology; heat loss; thermoregulation; UAV photogrammetry

Summary Statement

This study uses novel aerial photogrammetry methods to investigate how different-sized cetaceans have adapted to reduce heat loss.

1. Introduction

Endothermic species regulate their internal temperature by producing heat through natural metabolic processes (Irving 1973). Those species are able to adjust their heat production in response to environmental temperature variation, allowing them to maintain a relatively stable body temperature (i.e., homeothermy). Few endotherms live in a medium with ambient temperatures close to their body temperature, and most species that live in cold environments experience ambient temperatures towards the lower end of their thermoneutral zone (Davis 2019). Heat loss rate refers to the amount of heat energy transferred (lost) from an organism to its surroundings over a specific time period. The high thermal conductivity of water allows heat to pass through ~23 times faster than air, and therefore endotherms living in aquatic environments are subject to increased heat loss rates (Davis 2019). The higher rate of heat loss significantly elevates the metabolic costs of marine endotherms in their efforts to maintain homeothermy (Irving 1973), posing a considerable energetic challenge, especially to those inhabiting cold-water environments.

To facilitate life in a highly thermally conductive medium, the endothermic marine taxon of cetaceans has developed certain adaptations, such as a thick subdermal insulating layer of fat (i.e., blubber). Cetacean blubber is a complex tissue characterized by a collagen matrix, lipid-filled adipocytes, and vascular beds located near the epidermis (Parry 1949), and influences thermoregulation, energy storage, and locomotion (Noren and Wells 2009; Parry 1949; Struntz et al. 2004). The blubber tissue exhibits lower metabolic activity and requires less blood perfusion than most other tissue types (Sumich 1986), and therefore, serves as an efficient insulator. The structure, thickness, and thermal conductivity of blubber are key factors in minimizing heat loss in cetaceans (Dunkin et al. 2005; Kvadsheim, Folkow, and Blix 1994; McNab 2012; Worthy and Edwards 1990).

Cetaceans include lineages of the largest known animals to have ever existed on Earth (Fordyce and Marx 2018; Friedman et al. 2010; Gearty, McClain, and Payne 2018; Kelley and Pyenson 2015). Their large body size is advantageous in cold environments, as it offers lower surface-to-volume ratio (SVR), which leads to reduced rates of heat loss per unit of mass (Bergmann 1847; Blackburn, Gaston, and Loder 1999). A larger body size allows for efficient lipid storage, as well as adjustments in basal metabolic rate (BMR) and surface area (Brodie 1975). Interestingly, cetaceans also exhibit an exceptionally wide span of body sizes within the taxon, ranging from the < 1.5 m and < 45 kg vaquita (*Phocoena sinus* Norris & McFarland, 1958) to

the > 30 m and > 180 tons blue whale (*Balaenoptera musculus* Linnaeus, 1758). Despite this significant variation in size, the scaling of heat loss dynamics among different-sized species remains poorly understood, particularly in cases where smaller cetacean species co-exist with larger heterospecifics in the same cold-water ecoregion.

Heat loss estimations in cetaceans have been mainly based on Fourier's law of heat conduction and Kleiber's concept of a homeothermic animal (Kleiber 1932), wherein a heat-producing body core has a constant temperature and is surrounded by an insulating layer with a specified thermal conductivity, thickness, surface area and surface temperature. Only few studies have discussed the interspecific scaling of these parameters in cetaceans (Hokkanen 1990; Ryg et al. 1993). Specifically, (Ryg et al. 1993) discussed the interspecific scaling of insulation in marine mammals, including two cetacean species (harbour porpoises, *Phocoena phocoena* Linnaeus, 1758 and minke whales, *Balaenoptera acutorostrata* Lacépède, 1804). However, it is worth noting that the samples used in that study originated from different geographical areas, and thus did not account for environmental variation (i.e. differences in water temperature). Most other studies have investigated the thermal ecology of either large whales (Brodie 1975, 1981; Folkow and Blix 1992; Kshatriya and Blake 1988; Lavigne, Innes, and Edwards 1990; Parry 1949; Sumich 2021), or smaller cetaceans (Adamczak et al. 2020; Bagge et al. 2012; Hampton and Whittow 1976; McGinnis et al. 1972; Worthy and Edwards 1990).

Previous research on heat loss has been mainly based on samples that originate from purposely taken (Folkow and Blix 1992; Kvadsheim et al. 1994; Parry 1949; Ryg et al. 1993) or incidentally caught specimens (Ryg et al. 1993; Worthy and Edwards 1990), stranded animals (Adamczak et al. 2020; Bagge et al. 2012; Sumich 2021), captive animals (Hampton and Whittow 1976; McGinnis et al. 1972; Sumich 2021), or on previously published data used in theoretical models (Brodie 1975, 1981; Kshatriya and Blake 1988; Lavigne et al. 1990; Watts, Hansen, and Lavigne 1993). Relying on stranded animals may introduce biases, as these individuals might not represent a healthy population (Barratclough et al. 2014; Currie et al. 2021; Moore et al. 2004; Moore and Read 2008). Samples from incidentally caught cetaceans are more reliable, however these are mainly available for smaller individuals, since large whales do not usually suffer immediate drowning in the nets, but rather chronic entanglement in fishing gear (Moore and van der Hoop 2012). Using ultrasound to obtain non-invasive blubber thickness measurements is especially useful for captive animals or those in capture-release programs (Adamczak et al. 2021; Tang et al. 2023; Wells et al. 2004). For larger whales,

practical constraints limit non-invasive direct measurements from free-living animals, emphasizing the potential of utilizing historical whaling records. Empirical data from the whaling era have been proven to be highly valuable in supplementing our understanding of large whale physiology, especially when used in conjunction with non-invasive methods (Christiansen et al. 2019; Glarou et al. 2023).

Uncertainty pervades existing heat loss calculations, which are particularly hindered by inaccurate descriptions of cetacean body shapes, and subsequently, estimates of SVRs (Watts et al. 1993). Cetacean species exhibit significant variations in body shape, with some being stout, while others are more slender (Woodward, Winn, and Fish 2006). Most previous studies estimated the surface area of cetaceans using allometric equations (Folkow and Blix 1992; Kshatriya and Blake 1988; Lavigne et al. 1990; Parry 1949; Sumich 2021; Worthy and Edwards 1990), cylindrical or conical body shape models (Brodie 1975, 1981; Ryg et al. 1993; Worthy and Edwards 1990), and ellipsoid (prolate spheroid) models (Hokkanen 1990). While these models have been widely used, they fall short in capturing the elliptical cross-sectional body shape of cetaceans (Christiansen et al. 2019; Lockyer, McConnell, and Waters 1985). Recent methodological advancements, however, have substantially increased the accuracy of body shape models of cetaceans, by modelling their body as a series of infinitesimal ellipses using UAV (Unoccupied Aerial Vehicle) photogrammetry (Christiansen et al. 2019, 2020, 2021; Glarou et al. 2023) or by creating 3D models using specialized modelling software (Adamczak et al. 2019, 2020; Hirtle et al. 2022; Zhang et al. 2023).

The aim of this study was to compare how physiological thermoregulatory adaptations scale across five different-sized species of cetaceans living in the same cold-water ecoregion. To do that, we calculated the heat loss rates for each species, using a simplified heat loss model (Castellini et al. 2015; Folkow and Blix 1992; Strunk 1971; Sumich 2021), where the input parameters originated from: (i) body surface estimates from UAV images; (ii) blubber thickness predictions based on empirical blubber thickness measurements for a given length, taken from previously bycaught or caught specimens; (iii) published data on seawater temperature in NE-Iceland (Sigurðsson 2005), on body temperature of cetaceans (Hampton et al. 1971; Morrison 1962; Parry 1949; Vongraven et al. 1990; Yasui and Gaskin 1986) and on thermal conductivity of blubber (Dunkin et al. 2005; Folkow and Blix 1992; Parry 1949; Worthy and Edwards 1990). By incorporating body shape models derived from UAV photogrammetry, we aimed to improve the accuracy of the surface area estimates of the

different species, addressing an important limitation of traditional models. The general relationship between size and heat loss has not been previously examined in a comparative study of sympatric, different-sized cetaceans. Assessing how these different-sized species contend with heat loss in the same cold-water environment is crucial in understanding their overall energetic requirements, and ultimately, their vulnerability to climate-induced changes in their environment (IPCC 2014).

2. Materials and Methods

2.1. Data collection

2.1.1. Study site and target species

The waters surrounding Iceland are characterized by high productivity and cold temperatures (Astthorsson, Gislason, and Jonsson 2007; Sigurðsson 2005). The northern, subarctic Icelandic waters are influenced by two primary water masses: the relatively warm and saline North Atlantic waters (T = 6 – 11 ºC; S ~ 35.2 ppt) and the relatively fresh and very cold Arctic Ocean waters (T < 0 ºC; S < 34.5 ppt) (Astthorsson et al. 2007). The influx and mixing of the two water masses create harsh thermal conditions year-round. In North Iceland, the mean seawater temperature at 50 m depth was 3.69°C over a period of 53 years (Sigurðsson 2005).

Data collection was carried out in Skjálfandi Bay, Northeast Iceland (66°05′N, 17°19′W) between August 2021 and November 2023 (Figure S1). The bay extends approximately 30 km at its widest point and 20 km in length, with a total surface area of ~1100 km^2, while its deepest point is >150 m (Akamatsu, Rasmussen, and Iversen 2014). Skjálfandi Bay was a suitable study area for this study, as it constitutes an important feeding ground for several cetacean species (Akamatsu et al. 2014; Rasmussen 2009). Five cetacean species that are regular inhabitants of the bay were selected for the purpose of this study; from smallest to largest, these species are the: (i) harbour porpoises, (ii) white-beaked dolphins (*Lagenorhynchus albirostris* Gray, 1846), (iii) minke whales, (iv) humpback whales (*Megaptera novaeangliae* Borowski, 1781) and (v) blue whales.

2.1.2. UAV photogrammetry

Unoccupied Aerial Vehicles (UAVs) were used to capture aerial footage of the five target cetacean species in Skjálfandi Bay, NE-Iceland. Data collection was conducted over three years: (i) 2021 (from July 31st to November 16th - 20 days of sampling; Table 1); 2022 (from April 27th to November 15th - 48 days of sampling; Table 1); and 2023 (from April 2nd to November 19th - 71 days of sampling; Table 1). Sampling activities occurred from two vantage points: (i) mainly sea-based observations, employing a 5-meter research vessel to survey the bay and to operate the UAV; and (ii) opportunistic land-based UAV flights, with distances of up to 3 km from land, primarily from Húsavík Lighthouse, located at the south-eastern part of the study area (Figure S1).

We utilized four different UAV setups over the different field seasons (Table 1). In 2021, a DJI Inspire 1 Pro quadcopter (56 cm diameter, 3.4 kg, https://www.dji.com) was used, equipped with a 16-megapixel DJI Zenmuse X5 micro four-thirds camera with an Olympus M.Zuiko 25 mm f1.8 lens. Occasionally, a DJI Phantom 4 Pro+ v2.0 quadcopter (35 cm diameter, 1.368 kg, https://www.dji.com) was also used for data collection. In 2022 and 2023, the main UAV setup included a DJI Inspire 2 quadcopter (60 cm diameter, 3.4 kg, https://www.dji.com), equipped with a 20.8-megapixel Zenmuse X5S micro four-thirds camera with a Olympus M.Zuiko 25mm f/1.8 lens attached to it. Two additional UAV configurations were occasionally used during these two field seasons, mainly as backup systems: a DJI Phantom 4 Pro+ v2.0 quadcopter (35 cm diameter, 1.368 kg, https://www.dji.com) and a DJI Phantom 4 Advanced quadcopter (35 cm diameter, 1.368 kg, https://www.dji.com). In 2022 and 2023, a LightWare SF11/C laser rangefinder (Lightware Optoelectronics, weighing 35 g) was externally attached to the drones, to record accurate altitude measurements above sea level during the UAV flights.

Table 1. Summary table of sampling periods, UAV models and scaling method used in each sampling season.

Sampling period	UAV model	Scaling method
31st Jul 2021 – 16th Nov 2021	DJI Inspire 1 Pro & DJI Phantom 4 Pro+ v2.0	Known-sized object
27th Apr 2022 – 15th Nov 2022	DJI Inspire 2 & DJI Phantom 4 Pro+ v2.0	Laser rangefinder
2nd Apr 2023 – 19th Nov 2023	DJI Inspire 2 & DJI Phantom 4 Pro+ v2.0	Laser rangefinder

The UAVs were flown at an average altitude of 33.18 m (± 16.40 m). All UAVs were recording videos of the target species as they surfaced to breathe, in 4K resolution (3840 × 2160 pixels) with the camera facing down, stabilized at a 90-degree angle using a gimbal. To minimized radial distortion, the animals were consistently centred in the frame, where the effects of lens distortion are least pronounced (Burnett et al., 2019). Further lens distortion correction was omitted, as the effect of lens distortion of the Olympus M.Zuiko 25mm f/1.8 lens on measurement accuracy is minimal (1% error) (Dawson et al. 2017). The UAV was operated exclusively in favourable weather conditions, characterized by no precipitation and wind speeds below 15 knots.

2.2. UAV data processing

2.2.1. Morphometric measurements

During the post-processing stage, a still-frame photograph was extracted from each video recording, where the animal was lying flat at the water's surface, with its dorsal side clearly visible, no body arching, and a well-defined body contour (both in terms of length and width) (Figure S2; for details, see: Christiansen et al. 2016). In instances where the individual rolled over during video recording, additional still-frame photographs were extracted to capture the lateral side of the animal (Christiansen et al. 2019). Subsequently, each extracted photograph underwent a quality grading process, based on specific criteria, such as the camera focus, the degree of body roll, body arch and body pitch, as well as the accuracy of measurements for body length and body width (length and width measurability) (Christiansen et al. 2018). Quality grading was performed on a scale ranging from 1 to 3, where a grade of 1 indicated good quality, a grade of 2 indicated medium quality, and a grade of 3 indicated poor quality. Only the images that received a quality grade of 1 or 2, for both body length measurability (mean error of less than 0.38%) and body width measurability (a mean error of less than 2.31%), as well as the images with less than two grade 2 ratings regarding pitch, arch, and roll combined, were included in our subsequent analyses (Christiansen et al. 2018).

The selected images depicting the dorsal side of the animals were used to measure the total body length (BL, distance from tip of rostrum to the end of fluke notch) and the body width (W) at 5% intervals along the body axis (Christiansen et al. 2016). Analogously, images capturing the lateral side of the animals were used to measure the body height (H, the distance from the dorsal to ventral side) at the same 5% intervals as the width measurements

(Christiansen et al. 2019). All measurements (in pixels) were executed in R v4.0.5 (R Development Core Team 2018) using a custom-built R script (Christiansen et al. 2016).

Two different scaling methods were used to convert the pixel measurements to absolute measurements (in m), depending on the sampling year (Table 1). In 2021, the images were scaled using an object of known length (Christiansen et al. 2016; Glarou et al. 2023), whereas in 2022 and 2023, the altitude measurements from the laser rangefinder were used to scale the pixels to meters (for details, see: Christiansen et al. (2018)).

2.2.2. Body shape and volume

The three-dimensional body shape of the five species was described using an elliptical model that accurately captures the cross-sectional body shape of each species (Christiansen et al. 2019, 2020; Lockyer et al. 1985; Zhang et al. 2023). The height-to-width (HW) ratio at each 5% measurement site across the body was calculated for each species, using matching dorsal and lateral images (i.e., width and height measurements) (Christiansen et al. 2019). Hence, only the individuals for which both width and height measurements were available, were used to calculate the HW ratio and construct the body shape models. For the rest of the individuals, where only width measurements (dorsal photographs) were available, we derived the height measurements from the respective width measurements by applying the estimated HW ratio, specific to each measurement site.

We used measurements of body length (BL), width (BW), and height (H) to estimate body volume of each of the five target species. Firstly, the body model of the animals was split into 20 body segments and each segment corresponded to the section of the body between two adjacent width/height measurement sites. Each body segment (s) was then modelled as a series of infinitesimal ellipses and the volume of each segment (V_s) was calculated using the following equation (Equation 1; Christiansen et al. 2019):

$$V_{s,i} = BL_i \times 0.05 \times \int_0^1 \pi \times \frac{W_{A,s,i} + \left(W_{P,s,i} - W_{A,s,i}\right) \times x}{2} \times \frac{H_{A,s,i} + \left(H_{P,s,i} - H_{A,s,i}\right) \times x}{2} dx \quad (1)$$

where BL_i represents the body length of cetacean i, $W_{A,s,i}$ and $H_{A,s,i}$ pertain to the measurements of the anterior width and height of body segment s for individual i, and $W_{P,s,i}$ and $H_{P,s,I}$ refer to the measurements of the posterior width and height of segment s for individual i. To address

the gradual narrowing of the animal's body towards the extremities of the animal, we modelled the segments closest to the front (0-5% of BL) and those near the tail end (85-100% of BL) as elliptical cones (Christiansen et al. 2019).

Subsequently, the total body volume (V_{Tot}) for each individual cetacean (i) was estimated by summing the volumes of all the body segments (s) (Equation 2):

$$V_{Tot,i} = \sum_{s=1}^{20} V_{s,i} \quad (2)$$

2.3. Heat loss estimation

To estimate the rate of heat loss among the five target species, we used a simplified model that describes heat flow across a plane (Castellini et al. 2015; Folkow and Blix 1992; Strunk 1971; Sumich 2021). Key factors relevant to this model included: (i) the effective surface area of the body exposed to water; (ii) the insulation mechanisms, specifically the thickness and thermal conductivity of the blubber layer; (iii) and the temperature gradient between the cetacean's body and the surrounding aquatic environment.

To determine these variables, we employed a combination of: (i) surface area estimation derived from UAV measurements, (ii) physical measurements taken from previously bycaught or caught specimens (i.e., blubber thickness for a given body length), and published thermal conductivity of blubber (Dunkin et al. 2005; Folkow and Blix 1992; Parry 1949; Worthy and Edwards 1990), as well as (iii) published data on seawater temperature in NE-Iceland (Sigurðsson 2005) and on body temperature of cetaceans (Hampton et al. 1971; Morrison 1962; Parry 1949; Vongraven et al. 1990; Yasui and Gaskin 1986). These variables were inserted in Equation 3, which describes steady state heat flux from a body (Castellini et al. 2015; Folkow and Blix 1992; Strunk 1971; Sumich 2021):

$$HL = \frac{k \times SA \times (T_b - T_w)}{BT} \quad (3)$$

where heat loss (HL; in W) is a function of the thermal conductivity of blubber (k; in W m^{-1} $^oC^{-1}$), the blubber thickness (BT; in m), the thermally active body surface area (SA; in m^2), and the temperature gradient between internal body temperature (T_b; in ^oC) and ambient water temperature (T_w; in ^oC).

2.3.1. Surface Area

The surface area of each cetacean was determined from the UAV measurements, by employing a segmentation approach, where the body of each individual was divided into 1000 elliptical segments, with each ellipse representing a girth measurement on the body. We used the width and height measurements from each animal (divided by 2) as the major and minor radius of each ellipse. The width and height were interpolated between two consecutive measurement sites along the body. Using the complete elliptic integral of the second kind formula we were able to calculate the circumference (C) of each ellipse (i.e., girth segment, s) (Christiansen et al. 2019):

$$\boldsymbol{C_{s,i} = 4 \times \int_0^{\frac{\pi}{2}} \sqrt{r_{W,s,i}^2 \times \cos(x)^2 + r_{H,s,i}^2 \times \sin(x)^2}\, dx} \qquad (4)$$

where r_W and r_H represent the radius of the body width (W) and body height (H) of each girth segment s of individual i, respectively. Lastly, the total surface area was estimated by multiplying the circumferences of all elliptical segments with the total body length of each individual. This approach allowed us to accurately estimate the thermally active surface area of the entire body, while accounting for the varying shape along its length. Although flippers and flukes add to the overall body surface area, these body parts have counter-current vascular heat exchange systems (Adamczak et al. 2020; Scholander and Schevill 1955) that effectively minimize heat loss. Hence, these appendages were excluded when calculating the thermally active surface area (insulated by blubber) for the five target species.

2.3.2. Thermal conductivity of blubber

The thermal conductivity of blubber offers an absolute measure of heat transfer across its depth, and has been utilized, in conjunction with blubber thickness, to evaluate its insulative capability (Dunkin et al., 2005). Empirical species-specific blubber conductivity values in literature are very limited, and almost all originate from measurements on dead blubber tissues (Folkow and Blix 1992; Kvadsheim, Folkow, and Blix 1996; Parry 1949). Peripheral blood flow may affect the thermal conductivity of live blubber; however, this blood flow is minimal when an animal inhabits an environment with temperatures close to its lower critical temperature. Thus, it is safe to assume that the thermal conductivity of blubber in that environment would be close to that of dead blubber (Kvadsheim et al. 1996). This assumption has been validated in a study by Kanwisher and Sundnes (1965) where the blubber conductivity of a live harbour porpoise was measured and was found to be nearly equal to the thermal conductivity of dead blubber.

Worthy and Edwards (1990) reported a thermal conductivity value of 0.1 W m^{-1} $^{o}C^{-1}$ for harbour porpoise blubber. To our knowledge, the thermal conductivity of white-beaked dolphin blubber has not been examined. Therefore, we assumed a thermal conductivity of 0.18 W m^{-1} $^{o}C^{-1}$, which is the value measured for common bottlenose dolphins (*Tursiops truncatus* Montagu, 1821) (Dunkin et al. 2005). A value of 0.178 W m^{-1} $^{o}C^{-1}$ was used for minke whales (Folkow and Blix 1992), whereas for humpbacks and blue whales we used a thermal conductivity value of 0.2 W m^{-1} $^{o}C^{-1}$, which is the value calculated for large whales (Parry 1949).

2.3.3. Blubber thickness

For harbour porpoises, white-beaked dolphins and minke whales, blubber thickness measurements for a given body length were obtained from bycaught specimens in Icelandic waters between 1987 and 2009 (provided by the Marine & Freshwater Research Institute of Iceland (MFRI)). Additional measurements from harbour porpoises that were bycaught in Skjálfandi Bay, NE-Iceland between 2011-2012 were provided by the University of Iceland's Husavik Research Centre (UIHRC). For humpback whales, we used published data on the blubber thickness of whales caught in the North-West Atlantic (Eastern Canada) during 1969-1971 (Mitchell 1973). Lastly, historical whaling records of Soviet catches of blue whales from the Southern Ocean (between 1961-1975) were used to obtain blubber thickness values for this species. Even though these measurements were from another basin and therefore from a different population, we still chose to include them in this study, as no data from Iceland or the North Atlantic were available. Blubber thickness measurements for all species were taken from the flank at the midpoint of their lateral side below the dorsal fin (referred to as M4 measurement point) (Chosson et al. 2023; Christiansen et al. 2013; Lockyer and Waters 1986).

Linear models were constructed for each species to determine the relationship between blubber thickness and body length (BL_i), using the bycatch and historical whaling records, where blubber thickness was measured for a given length. Blubber thickness (BT_{rel}) was expressed as a proportion of body length (BL) of individual i, for each species.

$$\log(\boldsymbol{BT_{rel,i}}) = \boldsymbol{\alpha} + \boldsymbol{\beta} \times \log(\boldsymbol{BL_i}) + \boldsymbol{\varepsilon_i} \quad (\mathbf{5})$$

The blubber thickness of the free-living animals was predicted using two approaches: (i) based on the log-linear relationship between relative blubber thickness and body length of the species (Approach 1); and (ii) based on the mean relative blubber thickness of the species (Approach

2). In both cases, the predicted relative blubber thickness estimates were converted to absolute blubber thickness by multiplying the value with the total length of the animal. The absolute blubber thickness was then used as input on the heat flux equation (Equation 3) to estimate the heat loss of each individual in our dataset.

2.3.4. Temperature gradient

Mean seawater temperature at 50 m depth was measured to be 3.69 ^{o}C in North Iceland over the years 1952 – 2005 (Sigurðsson 2005), and therefore this value was set as the ambient water temperature (T_w) in Skjálfandi Bay. Existing data on species-specific body temperature are scarce. In most studies, body temperature measurements have been obtained post-mortem, from freshly-killed animals (Brodie and Paasche 1985; Morrison 1962; Parry 1949; Vongraven et al. 1990; Yasui and Gaskin 1986), while a few studies have also used telemetry to measure internal body temperature of smaller cetaceans in captivity (Hampton et al. 1971; McGinnis et al. 1972). Here, we used a body temperature of 37 ^{o}C for harbour porpoises (Yasui and Gaskin 1986), 37.3 ^{o}C for white-beaked dolphins (assumption based on the value for common bottlenose dolphins) (Hampton et al. 1971), 35.3 ^{o}C for minke whales (Vongraven et al. 1990), 36 ^{o}C for humpback whales (Morrison 1962) and 35.5 ^{o}C for blue whales (Parry 1949).

2.4. Partial effects plots and sensitivity analysis

To examine the impact of the individual model parameters on estimated heat loss, we used partial effects plots. Each plot varied a single parameter while maintaining constant values for all other variables, thereby isolating the influence of each parameter on heat loss. Four partial effects plots were created, one for each model parameter (i.e. surface area, body temperature, thermal conductivity and blubber thickness). The heat loss for each species was re-estimated for each plot, using the species-specific (i.e., varied) values of the parameter examined and the average values among all species as the constant values for all other parameters.

A sensitivity analysis was conducted to assess the robustness of between-species comparisons in heat loss estimates under different blubber thickness input scenarios. We used the upper and lower 95% confidence intervals (CI) to re-estimate heat loss for each species and compared these values with heat loss estimates derived from mean blubber thickness values (relative to BL) multiplied with each individual's BL. Using the two extreme blubber thickness values allowed us to evaluate the effect of seasonal variations in blubber thickness on the heat loss estimates of each species.

2.5. Model validation

Validation tests were conducted to verify that all necessary assumptions of the linear models were satisfied. The homogeneity of variance was explored with scatter plots of residuals against fitted values and against each explanatory variable. Additionally, we assessed the normality of residuals using quantile-quantile plots and residual histograms, and we checked for influential points and outliers using leverage and Cook's distance. All model assumptions were met.

3. Results

3.1. Sample size and effort

Body morphometric data on the five target species were collected on 139 days between 2021 and 2023 in Skjálfandi Bay, NE-Iceland (Table 2). In total, 929 still images of the target species were extracted in which the animals were displaying their dorsal side. Out of the 929 dorsal images, 685 images fulfilled the quality grading criteria. A total of 50 images of porpoises, 294 images of white-beaked dolphins, 30 images of minke whales, 282 images of humpback whales and 29 images of blue whales were included in the final analyses. In addition, 31 still images were selected, in which the target species displayed their lateral side. Specifically, we extracted 1 lateral image of harbour porpoises, 19 lateral images of white-beaked dolphins, 1 lateral image of minke whales, 9 lateral images of humpback whales and 1 lateral image of blue whales.

3.2. UAV morphometric measurements

The total length of harbour porpoises ranged from 0.97 to 1.60 m (mean = 1.28 ± 0.15), and their volume varied between 0.02 and 0.08 m^3 (mean = 0.04 ± 0.01) (Table 2). The anterior, cross-sectional body shape of the species is almost circular (up to 65% BL from the rostrum; Figure 1; Figure S3; Table S1), while towards the posterior end of their body, harbour porpoises exhibit a vertically elongated shape (Figure 1; Figure S3). Highest variation in body widths and heights (relative to BL) across the body of harbour porpoises was observed between 20% and 75% BL from the rostrum (Figure 1; Figure S3; Table S1). The mean head length of harbour porpoises was at approximately 8% of BL from the rostrum (Table S2).

The total length of white-beaked dolphins varied from 1.10 to 2.86 m (mean = 2.18 ± 0.45), while their body volume ranged between 0.02 and 0.38 m^3 (mean = 0.19 ± 0.09) (Table 2). The

cross-sectional shape of the white-beaked dolphin body is vertically elongated with HW ratios > 1 across the whole body, except for the anterior end, where their cross-section body shape is closer to circular (between 15-25% BL from the rostrum; Figure 1; Figure S3; Table S1). The vertically elongated shape is especially pronounced towards the posterior end of their body (from 70% BL from the rostrum; Figure 1; Figure S3; Table S1). The greatest variations in body widths and heights (relative to BL) among white-beaked dolphins were observed between 20% and 60% of the body length from the rostrum (Figure 1; Figure S3; Table S1). The mean head length of white-beaked dolphins was at approximately 11% of BL from the rostrum (Table S2).

The total length of minke whales ranged from 4.36 to 8.64 m (mean = 7.00 ± 1.17). Volume varied between 0.76 and 6.52 m^3 (mean = 3.49 ± 1.42) (Table 2). The cross-sectional body shape of minke whales is rounded for the most part (between 15% and 75% BL from the rostrum; Figure 1; Figure S3; Table S1), with the anterior and posterior ends being horizontally and vertically elongated, respectively. Body widths and heights (relative to BL) of minke whales showed the most variation around the middle part of their body, between 40% and 70% BL from the rostrum (Figure 1; Figure S3; Table S1). The mean head length of minke whales was approximately 18% of their BL (Table S2).

The total length of humpback whales varied from 6.01 to 14.20 m (mean = 9.39 ± 1.17). Volume ranged between 3.38 and 41.16 m^3 (mean = 11.23 ± 5.00) (Table 2). Humpback whales exhibit a mostly rounded cross-sectional body shape at the anterior and middle part of their bodies (between 15% and 60% BL from the rostrum; Figure 1; Figure S3; Table S1), while the posterior end of their body is elongated on the lateral plane. Similar to minke whales, the body widths and heights (relative to BL) of humpback whales varied noticeably around the middle part of their body, between 40% and 70% BL from the rostrum (Figure 1; Figure S3; Table S1). The mean head length of humpback whales was approximately 24% of their BL (Table S2).

The total length for blue whales ranged from 13.17 to 24.23 m (mean = 21.45 ± 2.17). Volume varied between 14.45 and 86.50 m^3 (mean = 64.16 ± 14.57) (Table 2). The anterior end of the cross-sectional body shape of blue whales is slightly flattened on the lateral plane, while their posterior end (from 50% BL from the rostrum; Figure 1; Figure S3; Table S1) is vertically elongated. Their shape at the middle part of the body is mostly circular (between 25% and 45% BL from the rostrum; Figure 1; Figure S3; Table S1), with HW ratios approaching 1:1. For

blue whales, highest variation in body widths and heights (relative to BL) was observed between 30% and 60% BL from the rostrum (Figure 1; Figure S3; Table S1). The mean head length of blue whales was approximately 21% of their BL (Table S2).

As expected, there was a strong positive relationship ($p < 0.001$, $R^2 > 0.85$) between body volume (BV) and body length (BL) on the log-log scale for all species.

3.3. Surface Area and Surface-Area-to-Volume Ratio (SVR)

The thermally active surface area of harbour porpoises ranged from 0.43 to 1.09 m^2 (mean = 0.71 ± 0.14; n = 50), with a SVR ranging from 13.85 to 21.72 (mean = 18.11 ± 1.61; n = 50). For white-beaked dolphins, the surface area varied from 0.48 to 3.39 m^2 (mean = 2.06 ± 0.71; n = 294), while the SVR from 8.50 to 24.75 (mean = 12.14 ± 2.96; n = 294) (Table 2). Minke whales had a surface area between 5.77 and 23.90 m^2 (mean = 15.54 ± 4.65; n = 30), and SVR between 3.67 and 7.70 (mean = 4.89 ± 1.16; n = 30). The surface area of humpback whales ranged from 14.32 to 78.06 m^2 (mean = 32.94 ± 8.84; n = 282), while their SVR was between 1.87 and 4.25 (mean = 3.09 ± 0.39; n = 282) (Table 2). Lastly, the surface area of blue whales varied from 46.51 to 150.55 m^2 (mean = 123.57 ± 20.85; n = 29), and their SVR was between 1.74 and 3.22 (mean = 1.99 ± 0.29; n = 29) (Table 2). SVR decreased with increasing body length for all species (Figure 2).

Table 2. Summary table showing the morphometric measurements deriving from UAV images for all target species.

Species	**N**		***Length (m)***	***Volume (m^3)***	***Surface Area (m^2)***	***SVR***
P. phocoena	*50*	*mean ± SD*	1.28 ± 0.15	0.04 ± 0.01	0.71 ± 0.14	18.11 ± 1.61
		min – max	0.97 - 1.60	0.02 - 0.08	0.43 - 1.09	13.85 - 21.72
L. albirostris	*294*	*mean ± SD*	2.18 ± 0.45	0.19 ± 0.09	2.06 ± 0.71	12.14 ± 2.96
		min – max	1.10 - 2.86	0.02 - 0.38	0.48 - 3.39	8.50 – 24.75
B. acutorostrata	*30*	*mean ± SD*	7.00 ± 1.17	3.49 ± 1.42	15.54 ± 4.65	4.89 ± 1.16
		min – max	4.36 - 8.64	0.76 - 6.52	5.77 - 23.90	3.67 - 7.70
M. novaeangliae	*282*	*mean ± SD*	9.39 ± 1.17	11.23 ± 5.00	32.94 ± 8.84	3.09 ± 0.39
		min – max	6.01 - 14.20	3.38 - 41.16	14.32 - 78.06	1.87 - 4.25

B. musculus	*29*	*mean ± SD*	21.45 ± 2.17	64.16 ± 14.57	123.57 ± 20.85	1.99 ± 0.29
		min – max	13.17 - 24.23	14.45 - 86.50	46.51 - 150.55	1.74 - 3.22

3.4. Blubber thickness predictions

Using the blubber thickness measurements that originated from the different data sources (bycaught and caught animals), we found a significant relationship between blubber thickness (relative to BL) and body length (BL) on the log-log scale for harbour porpoises ($F_{1,498} = 394$, $p < 0.001$), white-beaked dolphins ($F_{1,39} = 31.48$, $p < 0.001$), minke whales ($F_{1,14} = 19.56$, $p < 0.001$) and blue whales ($F_{1,331} = 49.57$, $p < 0.001$); for humpback whales the relationship was not significant ($F_{1,34} = 1.523$, $p = 0.23$). Therefore, we used the linear models to predict the blubber thickness of the UAV-derived samples from their estimated length (Approach 1), and subsequently used those predictions as input variables in our heat loss model. For comparison, heat loss was re-estimated using blubber thickness predictions derived from mean relative blubber thickness value (relative to BL) for each species (Approach 2). The effect of BL on blubber thickness, and subsequently on heat loss, was evident in all species (Figure S4). Thus, blubber thickness predictions from Approach 1 were deemed more accurate and were used in the final heat loss models.

The predicted blubber thickness (Approach 1) of the individuals measured with the UAV ranged from 16 to 23 mm (mean = 21 ± 2 mm; n = 50) for harbour porpoises, 14 to 18 mm (mean = 17 ± 1 mm; n = 294) for white-beaked dolphins, 33 to 40 mm (mean = 38 ± 2 mm; n = 30) for minke whales, 60 to 106 mm (mean = 84 ± 7 mm; n = 282) for humpback whales, and 45 to 114 mm (mean = 95 ± 14 mm; n = 29) for blue whales (Table 3). On an interspecific level, the blubber thickness (relative to BL) scaled negatively with increasing species size, except for humpback whales, which had relatively thicker blubber for a given length compared to minke whales and blue whales (Figure 3).

3.5. Heat loss estimation

3.5.1. Species-specific heat loss estimations

Harbour porpoises, the smallest cetacean of this study, exhibited a heat loss rate ranging from 63 to 210 W (mean = 113 ± 33 W), while their volume-specific heat loss rates spanned from 2471 to 3303 W m^{-3} (mean = 2854 ± 179 W m^{-3}) (Figure 4; Table 3). White-beaked dolphins

had a heat loss rate between 211 and 1,366 W (745 ± 259 W), and notably higher volume-specific heat loss, ranging from 3,075 to 10,878 W m^{-3} (4,451 ± 1,338 W m^{-3}) (Figure 4; Table 3). The heat loss rate for minke whales was estimated between 971 and 3543 W (mean = 2247 ± 631 W), with volume-specific heat loss spanning from 543 to 1,279 W m^{-3} (725 ± 222 W m^{-3}) (Figure 4; Table 3). Humpback whales exhibited heat loss rates from 1,535 to 4,776 W (2,508 ± 455 W), while their volume-specific heat loss ranged from 116 to 454 W m^{-3} (242 ± 48 W m^{-3}) (Figure 4; Table 3). Lastly, blue whales had the highest absolute heat loss rates between 6,615 and 9,173 W (8,274 ± 506 W), while also having the lowest average volume-specific heat loss rate, ranging from 100 to 458 W m^{-3} (143 ± 69 W m^{-3}) (Figure 4; Table 3).

Table 3. Summary table with the species-specific parameters used in the heat loss model, as well as the resulting heat loss (absolute) and volume-specific heat loss (relative) estimates for each species. n = sample number; Tw = Water temperature; Tb = Body temperature; k = thermal conductivity of blubber; SA = Surface area; BT = Blubber thickness; HL = Heat loss rate.

Species	*n*	T_w	T_b	*k*	*SA*	*BT (m)*	*HL (W)*	*HL (W m^{-3})*	
P. phocoena	*50*	3.69	37	0.1	0.71 ± 0.14	0.021 ± 0.002	113 ± 33	2,854 ± 179	*mean ± SD*
					0.43 – 1.09	0.016 – 0.023	63 – 210	2,471 – 3,303	*min – max*
L. albirostris	*294*	3.69	37.3	0.18	2.06 ± 0.71	0.017 ± 0.001	745 ± 259	4,451 ± 1,338	*mean ± SD*
					0.48 – 3.39	0.014 – 0.018	211 – 1,366	3,075 – 10,878	*min – max*
B. acutorostrata	*30*	3.69	35.3	0.178	15.54 ± 4.65	0.038 ± 0.002	2,247 ± 631	725 ± 222	*mean ± SD*
					5.77 – 23.90	0.033 – 0.040	971 – 3,543	543 – 1,279	*min – max*
M. novaeangliae	*282*	3.69	36	0.2	32.94 ± 8.84	0.084 ± 0.007	2,508 ± 455	242 ± 48	*mean ± SD*
					14.32 – 78.06	0.060 – 0.106	1,535 – 4,776	116 – 454	*min – max*
B. musculus	*29*	3.69	35.5	0.2	123.57 ± 20.85	0.095 ± 0.014	8,274 ± 506	143 ± 69	*mean ± SD*
					46.51 – 150.55	0.045 – 0.114	6,615 – 9,173	100 – 458	*min – max*

3.5.2. Effect of input parameters on heat loss estimations

To determine the effect of the input parameters on the heat loss estimations, partial effects plots were created (Figure 5). Relative blubber thickness and blubber thermal conductivity seemed

CONFIDENTIAL

to affect heat loss estimates the most (Figure 5C, 5D, 5E, 5F), whereas SA and body temperature have a smaller effect on heat loss estimations (Figure 5A, 5B, 5G, 5H).

The effect of seasonal variations in blubber thickness was examined through a sensitivity analysis, where we used the upper and lower 95% CI for blubber thickness to re-estimate the heat loss for all species, signifying the two extreme seasonal periods with the thickest and thinnest blubber layer, respectively. The results showed that the interspecific comparative relationships remained valid in both extreme scenarios (Figure 6).

4. Discussion

This study presents a comparative investigation of the different parameters that affect heat loss in five cetacean species of varying body size inhabiting the same cold-water marine environment, thus providing novel insights into their size-dependent thermoregulatory strategies. While we utilized a simplified heat loss model that has been used in previous studies (Castellini et al. 2015; Folkow and Blix 1992; Strunk 1971; Sumich 2021) and existing physiological data on body temperature, blubber thickness and thermal conductivity of blubber tissue, the surface area data were obtained using novel UAV imagery and aerial photogrammetry methods applied to free-living animals. To our knowledge, this is the first study that incorporates UAV photogrammetry methods to inform heat loss in cetaceans. Our UAV approach improved the accuracy of surface area estimations, compared to traditional methods, which in turn refined the heat loss calculations.

Traditional methods for surface area estimation of cetaceans have included allometric equations (Folkow and Blix 1992; Kshatriya and Blake 1988; Lavigne et al. 1990; Parry 1949; Sumich 2021; Worthy and Edwards 1990), or modelling their body as a cylinder, cone or ellipsoid (prolate spheroid) (Brodie 1975, 1981; Hokkanen 1990; Ryg et al. 1993; Worthy and Edwards 1990). However, these models do not capture the true variations in the body shape of various cetacean species (Christiansen et al. 2019; Lockyer et al. 1985), and thus may result in inaccurate estimates of their surface area. In this study, we were able to improve the accuracy of surface area estimation by modelling the body shape of each species as a series of infinitesimal ellipses using previously established UAV photogrammetry methods (Christiansen et al. 2019, 2020, 2021; Glarou et al. 2023). The toothed whales (i.e., harbour porpoises and white-beaked dolphins) were more similar in shape, having a more rounded, robust body that tapers towards the fluke, while the baleen whales (i.e., minke, humpback, and

blue whales) all had a fusiform shape and similar HW ratios across their body (Figure 1; Figure S3). SVR decreased with increasing body length consistently across all species (Figure 2), reflecting the cubic increase in body volume with length compared to the quadratic increase in surface area with length. Lower SVR means that there is less exposed body surface area through which heat can escape, which in theory should result in lower heat loss. This was confirmed through our heat loss model, which showed that cetaceans with lower SVR had lower volume-specific heat loss estimates, highlighting the effect of body size on heat loss both in an intraspecific and interspecific level (Figure 2; Figure 4).

Only a handful of studies have attempted to estimate heat loss rates for cetaceans (Bagge et al. 2012; Brodie 1975; Folkow and Blix 1992; Kshatriya and Blake 1988; Lavigne et al. 1990; Meagher et al. 2008; Parry 1949; Sumich 2021). Regarding our target species, specific heat loss calculations are only available for harbour porpoises, minke whales, and blue whales (Folkow and Blix 1992; Kshatriya and Blake 1988; Lavigne et al. 1990; Parry 1949), while, to our knowledge, no information exists for white-beaked dolphins and humpback whales. Parry (1949) reported an average heat loss of 302 W m^{-2} (converted from the provided value of 260 kcal/m^2/h) for a 155 cm harbour porpoise living in 10° C waters. For comparison, the average heat loss rate per m^2 for harbour porpoises in our study was lower, at 158.4 W m^{-2}. Although the temperature gradient was greater in our study area, the lower heat loss values predicted from our model are likely more accurate, as Parry (1949) used the thermal conductivity values for fin whale (*Balaenoptera physalus* Linnaeus, 1758) blubber for their calculations. Later experiments revealed that the blubber of harbour porpoises is much less conductive than other cetaceans, and hence a more effective insulator (Worthy and Edwards 1990). The average rate of sensible heat loss for minke whales in cold waters has been reported to be 3.81 ± 0.53 W $kg^{-0.75}$ (SD = 0.53) (Folkow and Blix 1992). The mass-specific values provided by Folkow and Blix (1992) can be converted to absolute heat loss rates for the purpose of comparison with our estimations, resulting in absolute heat loss rates between 1,205 and 2,130 W (mean = 1,700 ± 340 W), which are slightly lower than our predictions for similarly sized animals (range = 971 – 3,543 W; mean = 2,247 ± 631 W). This difference may be attributed to a steeper temperature gradient, as the empirical calculations of heat loss for minke whales referred to slightly higher water temperatures (mean = 4.2 ± 1.1 ^{o}C) than our study site, as well as lower body temperatures of individual whales (mean = 28.8 ± 1.7 ^{o}C) (Folkow and Blix 1992). Lastly, various theoretical models have been used in the past to predict the heat loss of blue whales (Kshatriya and Blake 1988; Lavigne et al. 1990), however the resulting theoretical predictions are higher than our

findings. Kshatriya and Blake (1988)'s model included erroneous assumptions that lead to severely overestimated heat loss rates, and subsequently, very high estimates of lowest critical temperatures, as discussed in Lavigne et al. (1990). Although Lavigne et al. (1990) corrected many of the inaccuracies reported in Kshatriya and Blake (1988), they still utilized theoretical values for body size and shape, and allometric equations to predict most input variables of the heat loss model, and therefore, their results are inherently less reliable.

We found a significant negative relationship between blubber thickness (relative to BL) and BL for all species, except humpback whales, indicating that the relative thickness of blubber declined with increasing body length within a species. Consequently, the species-specific relationships between BL and blubber thickness were used to predict blubber thickness for all individuals in this study, which was ultimately incorporated in our final heat loss estimations. Although we did not find a significant relationship between the two variables for humpback whales, we still chose to predict blubber thickness from body length, as the observed non-significant relationship may have been due to the lack of younger (i.e. smaller) animals in the historical whaling (BL = 9.72 - 15.1 m; mean = 13.1 ± 1.12 m) dataset (Mitchell 1973). Juvenile humpback whales have been reported to have thicker blubber than adults (Holyoake, Stephens, and Coughran 2012), and therefore we assumed that using the mean blubber thickness (relative to BL) to estimate blubber thickness would be less accurate than using our linear model between blubber thickness and body length (Figure S4).

The partial effects plots showed that blubber thickness had the most influence on the heat loss estimations, followed by the thermal conductivity of blubber (Figure 5C, 5D, 5E, 5F), while SA and body temperature seemed to have a smaller effect (Figure 5A, 5B, 5G, 5H). The strong effect of blubber thickness on heat loss is corroborated by previous research in which various heat loss models were highly sensitive to blubber thickness variations (Watts et al. 1993). For most cetaceans, blubber thickness varies temporally both within and between years, as well as spatially across the body (Christiansen et al. 2013; Evans, Hindell, and Thiele 2003; Gunnlaugsson et al. 2020; Koopman 1998; Mackintosh, Wheeler, and Clowes 1929; Meagher et al. 2008; Miller et al. 2011; Rojano-Donãte et al. 2018; Siebert et al. 2022; Víkingsson 1990). In this study, we used blubber thickness data from various sources, where the animals were sampled at different times of the year and at different measurement points, and hence, the measured blubber thickness was affected by the time of year the sample was taken and by the topographic location across the body. Seasonal fluctuations in morphological body condition

(i.e., variations in blubber thickness) were also captured by the UAV measurements, as they were taken at various points over the year (Table 1). The purpose of this study, however, was to investigate the interspecific patterns of heat loss and not the within-species variations; our sensitivity analysis showed that the interspecific relationship remained relatively constant, even when using the extreme lower and upper 95% CI of the blubber thickness values in our model predictions (Figure 6). Thus, our comparative interspecific analyses were robust to any temporal biases introduced by the various datasets used in this study. Any topographical variation in blubber thickness was minimized by using the standardized M4 measurement point for all species, which was the only location where blubber thickness was measured in all the different datasets.

Despite our efforts to ensure accuracy in our estimations, certain factors introduced uncertainty into our heat loss model. A necessary caveat was the use of blubber thickness datasets from non-Icelandic populations of humpback whales and blue whales, which may be morphologically different. For example, Barlow et al. (2023) found morphological differences among three different populations of blue whales that could be attributed to different environmental conditions, feeding behaviours and migration strategies. Further limitations stem from the scarcity of species-specific information on thermal conductivity. While thermal conductivity values exist for some delphinid species, no data was available for white-beaked dolphins. Instead, we utilized published thermal conductivity values for bottlenose dolphins, the only dolphin species with documented blubber conductivity that inhabits regions beyond the tropics (Dunkin et al. 2005). For large baleen whales, the only available thermal conductivity value is that of fin whale blubber (Parry 1949). Thus, we inferred that both humpback and blue whales had similar insulative qualities in their blubber, an assumption which potentially increased the uncertainty in our estimations. Moreover, thermal conductivity measurements originated from dead blubber tissue, excluding the effect of peripheral perfusion which effectively allows an animal to change their conductance by increasing or decreasing the blood flow to the blubber (Favilla, Horning, and Costa 2022). Lastly, our calculations referred only to heat loss through the surface area of the body and excluded respiratory heat loss, which has been estimated to comprise about 6% of total heat loss in minke whales (Folkow and Blix 1992).

Despite the uncertainties listed above, our findings provide novel insights into the thermal physiology and heat loss scaling among the target species. The effect of blubber thickness and

its thermal conductivity on heat loss was particularly evident in harbour porpoises, where smaller (younger) individuals were able to offset their higher SVR by having a thicker relative blubber layer. In addition, harbour porpoise blubber had the lowest thermal conductivity of all species examined in this study. This is in line with prior research reporting that the quantity and quality of harbour porpoise blubber can yield up to four times the insulative capacity of other small cetaceans (Worthy and Edwards 1990). Furthermore, our findings suggest that white-beaked dolphins experience higher volume-specific heat loss, when compared to similarly-sized harbour porpoises. A previous study found that the thermal conductivity of blubber varies among the different reproductive classes in bottlenose dolphins, with adult individuals having higher thermal conductivity compared to sub-adults (Dunkin et al. 2005). If the same holds true for white-beaked dolphins, the lower thermal conductivity of blubber in sub-adults could partially offset the observed elevated heat loss in younger dolphins (Figure 4). White-beaked dolphins can be found year-round in Icelandic waters (Magnúsdóttir 2007), although their sighting rates are usually higher during the summers in coastal waters (Kristensen 2023), which coincides with the start of the calving period (Galatius and Kinze 2016). This seasonal pattern coupled with the high heat loss rates of smaller white-beaked dolphins observed in this study, suggests that the timing of parturition may be influenced by optimal thermal conditions for calf development. While uncertainty in thermal conductivity may affect our estimates, blubber thickness remains the predominant factor influencing heat loss, for which we had reliable data from Iceland for white-beaked dolphins. Consequently, it appears that white-beaked dolphins may indeed be less effective in reducing heat loss compared to similar-sized porpoises; in that scenario, a rapid somatic growth in early-life stages would be highly beneficial for young white-beaked dolphins to reduce their SVR to offset the higher heat loss. Additional adaptations may include higher metabolic rates and prey consumption to compensate for the elevated energetic cost of homeothermy, as reported for other smaller cetaceans, such as harbour porpoises (Rojano-Donãte et al. 2018; Wisniewska et al. 2016).

The larger baleen whale species in this study had a relatively thinner blubber layer (relative to BL) compared to the smaller odontocetes (Figure 3), while simultaneously having lower volume-specific heat loss rates (Figure 4). This is a fundamental consequence of scaling, where larger-bodied cetaceans require proportionally less blubber to achieve the same level of absolute insulation per unit of body surface area due to their lower SVRs (Goldbogen, Pyenson, and Madsen 2023). The advantage of a larger body size was particularly evident in blue whales, which exhibited the lowest volume-specific heat loss due to their low SVR despite having the

lowest relative blubber thickness among all species in this study. This indicates that blue whales are able to offset heat loss simply by being large. Interestingly, humpback whales had a lower heat loss compared to both blue whales and minke whales over the range of body lengths where the species overlapped. This may be attributed to differences in body shape (and consequently in SVRs), with humpback whales being more rotund (having relatively higher width and height measurements) and having relatively thicker blubber compared to the other baleen whales in this study (Figure 2, 3). Such differences are likely dictated by the life-history traits of each species (Woodward et al. 2006), and reflect trade-offs between thermoregulatory needs, energy storage, and locomotion. For instance, minke whales and blue whales are fast swimmers, benefiting from a slender and streamlined body that allows for efficient locomotion (Bannister 2018; Woodward et al. 2006), and thus, any excess blubber could potentially impede their swimming performance. While all three baleen whale species are migratory capital breeders, humpback whales undertake the longest known mammalian migration between feeding and breeding grounds (Kettemer et al. 2022; Robbins et al. 2011; Stevick et al. 2011), potentially incurring higher energetic costs. During this fasting period, humpback whales rely primarily on stored energy (in the form of excess blubber) to finance their reproduction (Stephens et al. 2009), thus having extra blubber can prove particularly advantageous for this species.

Our results show that cetaceans have developed divergent, size-dependent strategies to mitigate heat loss. Specifically, we found that smaller cetaceans rely on physiological and morphological adaptations (i.e. blubber quality and quantity) to reduce heat loss, whereas larger species rely mainly on their larger body size and lower SVR. Overall, larger mysticetes had significantly lower volume-specific heat loss compared to smaller odontocetes, which in turn reduces their mass-specific metabolic costs. This explains the remarkable fasting capacity of larger whales (Millar and Hickling 1990), which allows them to undertake long migrations and exploit a wide range of habitats; this is reflected in the seasonal occurrence of baleen whales in Iceland (Bacon et al. 2023; Basran et al. 2023; Heide-Jørgensen et al. 2001; Víkingsson and Heide-Jørgensen 2015). In contrast, smaller cetaceans will need to feed continuously to cover their high mass-specific metabolic costs, and thus, are more constrained to habitats with abundant and accessible prey (Kanwisher and Sundnes 1965; Wisniewska et al. 2016). Since blubber thickness is directly linked to prey intake (Rojano-Donãte et al. 2018), smaller cetaceans, which heavily rely on their blubber for insulation, may be particularly sensitive to climatic changes that impact prey availability. Understanding the implications of these size-dependent thermoregulatory strategies for prey requirements and habitat use is crucial to assess

the vulnerability of these species to climate-induced environmental changes, ultimately informing management decisions.

Acknowledgements

The authors would like to thank captain Aðalgeir Bjarnason, Charla Basran, Yakamoz Kizildas, and the students of the Húsavík Research Centre for their assistance during fieldwork. We extend our gratitude to Romane Renou, Francesco Manzi, Laura Monderde, and Carolina Di Napoli for their help with data processing. We also thank Christian Schmidt and the local whale-watching companies for keeping us updated about ongoing cetacean sightings during all field seasons. We are grateful to Vasileios Glaros for voluntarily building one of the laser rangefinder setups used in this study and for his technical support on equipment issues. Lastly, we would like to thank Whale Wise for lending us their Phantom 4 Advanced UAV setup, which allowed for the continuation of data collection in 2022 when the other UAV setups were malfunctioning.

Competing Interests

All authors declare no conflict of interest.

Funding

M.G. received funding from the University of Iceland's Doctoral Grants Fund (The Eimskip University Fund/Háskólasjóður Eimskipafélags Íslands) and the Company of Biologists Travelling Fellowship. A.P, S.N.S.R, D.S. and M.J. received support from the Erasmus+ Programme for Traineeships during data collection. M.H.R. received funding from the University of Iceland Research Fund to cover the fieldwork expenses. The sampling of bycaught animals was funded by the Marine and Freshwater Research Institute.

Author Contributions

M.G., F.C. and M.H.R. conceived the study; M.G. and F.C. designed methodology; M.G., A.P., S.N.S.R., D.S. and M.J. collected the UAV data; M.R.I, G.M.S., S.D.H., V.C., P.J.C., Y.I., D.T. collected/provided the blubber thickness data; M.G., A.P., S.N.S.R., D.S., Z.S. and M.J. processed the UAV data; M.G. conducted the analysis with input from F.C.; M.G. led the writing of the manuscript with input from F.C. and M.H.R.. All authors contributed critically to the drafts and gave final approval for publication.

CONFIDENTIAL

Data availability

The data that support the results of this study will be made available by the corresponding author upon request.

Diversity and Inclusion Statement

Our study involves authors from multiple countries, including scientists based in Iceland, the country where the study was conducted. All authors were engaged early on to ensure that diverse perspectives were considered. Relevant literature published by local scientists was cited whenever applicable.

References

Adamczak, S. K., Ann Pabst, W. A. McLellan, and L. H. Thorne. 2019. "Using 3D Models to Improve Estimates of Marine Mammal Size and External Morphology." *Frontiers in Marine Science* 6(JUN):1–12. doi: 10.3389/fmars.2019.00334.

Adamczak, Stephanie K., Rachel R. Holser, Daniel P. Costa, Elizabeth J. Berens McCabe, and Randall S. Wells. 2021. "Body Composition of Common Bottlenose Dolphins in Sarasota Bay, Florida." *Frontiers in Marine Science* 8. doi: 10.3389/fmars.2021.615773.

Adamczak, Stephanie K., D. Ann Pabst, William A. McLellan, and Lesley H. Thorne. 2020. "Do Bigger Bodies Require Bigger Radiators? Insights into Thermal Ecology from Closely Related Marine Mammal Species and Implications for Ecogeographic Rules." *Journal of Biogeography* 47(5):1193–1206. doi: 10.1111/jbi.13796.

Akamatsu, Tomonari, Marianne Helene Rasmussen, and Maria Iversen. 2014. "Acoustically Invisible Feeding Blue Whales in Northern Icelandic Waters." *The Journal of the Acoustical Society of America* 136(2):939–44. doi: 10.1121/1.4887439.

Víkingsson, Gísli A. 1990. "Energetic Studies on Fin and Sei Whales Caught off Iceland." *Report International Whaling Commission* 40(1):365–73.

Astthorsson, Olafur S., Astthor Gislason, and Steingrimur Jonsson. 2007. "Climate Variability and the Icelandic Marine Ecosystem." *Deep-Sea Research Part II:*

CONFIDENTIAL

Topical Studies in Oceanography 54(23–26):2456–77. doi: 10.1016/j.dsr2.2007.07.030.

Bacon, Cathy E., Katharine A. Hart, Ted Cheeseman, Lee Munson, Marianne H. Rasmussen, Valerie Chosson, and Charla J. Basran. 2023. "Short Communication: First-Documented Humpback Whale (Megaptera Novaeangliae) Photo-Identification Match and Round-Trip Migration between Iceland and the Turks and Caicos Islands." *J. Cetacean Res. Manage.* 24(1):161–68. doi: 10.47536/jcrm.v24i1.405.

Bagge, Laura E., Heather N. Koopman, Sentiel A. Rommel, William A. McLellan, and D. A. Pabst. 2012. "Lipid Class and Depth-Specific Thermal Properties in the Blubber of Two Species of Odontocete Cetaceans, the Short-Finned Pilot Whale (*Globicephala Macrorhynchus*) and the Pygmy Sperm Whale (*Kogia Breviceps*)." *Journal of Experimental Biology* 215(24):4330–39. doi: 10.1242/jeb.071530.

Bannister, John L. 2018. "Baleen Whales (Mysticeti)." Pp. 62–69 in *Encyclopedia of Marine Mammals*. Elsevier.

Barlow, D. R., K. C. Bierlich, W. K. Oestreich, G. Chiang, J. W. Durban, J. A. Goldbogen, D. W. Johnston, M. S. Leslie, M. J. Moore, J. P. Ryan, and L. G. Torres. 2023. "Shaped by Their Environment: Variation in Blue Whale Morphology across Three Productive Coastal Ecosystems." *Integrative Organismal Biology* 5(1). doi: 10.1093/iob/obad039.

Barratclough, Ashley, Paul D. Jepson, Philip K. Hamilton, Carolyn A. Miller, Kenady Wilson, and Michael J. Moore. 2014. "How Much Does a Swimming, Underweight, Entangled Right Whale (Eubalaena Glacialis) Weigh? Calculating the Weight at Sea, to Facilitate Accurate Dosing of Sedatives to Enable Disentanglement." *Marine Mammal Science* 30(4):1589–99. doi: 10.1111/mms.12132.

Basran, Charla, Valerie Chosson, Andre Williams, Nikola Simpson, Syndi Anne Long, Flordespina Dodds, Marianne H. Rasmussen, and Julia A. Horrocks. 2023. "Short Communication: First Documented Migration of an Icelandic Humpback Whale Mother and Calf Pair from the West Indies Breeding Grounds." *J. Cetacean Res. Manage.* 24(1). doi: 10.47536/jcrm.v24i1.833.

CONFIDENTIAL

Bergmann, Carl. 1847. “Über Die Verhältnisse Der Wärmeökonomie Der Thiere Zu Ihrer Größe.” *Göttinger Studien* 3:595-708.

Blackburn, Tim M., Kevin J. Gaston, and Natasha Loder. 1999. “Geographic Gradients in Body Size: A Clarification of Bergmann’s Rule.” *Diversity and Distributions* 5(4):165–74. doi: 10.1046/j.1472-4642.1999.00046.x.

Brodie, P. F. 1975. “Cetacean Energetics, an Overview of Intraspecific Size Variation.” *Ecology* 56(1):152–61.

Brodie, P. F. 1981. “A Preliminary Investigation of the Energetics of the Bowhead Whale (Balaena Mysticetus L).” *Rep. Int. Whal. Commn 31*.

Brodie, Paul, and Arvid Paasche. 1985. “Thermoregulation and Energetics of Fin and Sei Whales Based on Postmortem, Stratified Temperature Measurements.” *Canadian Journal of Zoology* 63(10):2267–69. doi: 10.1139/z85-336.

Castellini, Michael A, Mellish, and Jo-Ann. 2015. *Marine Mammal Physiology*. edited by M. A. Castellini and J.-A. Mellish. CRC Press.

Chosson, Valérie, Haseeb S. Randhawa, Guðjón M. Sigurðsson, Sverrir D. Halldórsson, Þorvaldur Björnsson, Vilhjálmur Svansson, Sandra M. Granquist, Karl Gunnarsson, Filipa I. P. Samarra, and Christophe Pampoulie. 2023. “First Record of Risso’s Dolphin Grampus Griseus (Cuvier, 1812) in Icelandic Waters.” *Ecology and Evolution* 13(9). doi: 10.1002/ece3.10477.

Christiansen, F., F. Rodríguez-González, S. Martínez-Aguilar, J. Urbán, S. Swartz, H. Warick, F. Vivier, and L. Bejder. 2021. “Poor Body Condition Associated with an Unusual Mortality Event in Gray Whales.” *Marine Ecology Progress Series* 658:237–52. doi: 10.3354/meps13585.

Christiansen, F., Fabien Vivier, Claire Charlton, Rhianne Ward, Alicia Amerson, Stephen Burnell, and Lars Bejder. 2018. “Maternal Body Size and Condition Determine Calf Growth Rates in Southern Right Whales.” *Marine Ecology Progress Series* 592:267–82. doi: 10.3354/meps12522.

Christiansen, Fredrik, Antoine M. Dujon, Kate R. Sprogis, John P. Y. Arnould, and Lars

CONFIDENTIAL

Bejder. 2016. “Noninvasive Unmanned Aerial Vehicle Provides Estimates of the Energetic Cost of Reproduction in Humpback Whales.” *Ecosphere* 7(10):1–18. doi: 10.1002/ecs2.1468.

Christiansen, Fredrik, Mariano Sironi, Michael J. Moore, Matías Di Martino, Marcos Ricciardi, Hunter A. Warick, Duncan J. Irschick, Robert Gutierrez, and Marcela M. Uhart. 2019. “Estimating Body Mass of Free-Living Whales Using Aerial Photogrammetry and 3D Volumetrics.” *Methods in Ecology and Evolution* 10(12):2034–44. doi: 10.1111/2041-210X.13298.

Christiansen, Fredrik, Kate R. Sprogis, Jasmin Gross, Juliana Castrillon, Hunter A. Warick, Eva Leunissen, and Susan Bengtson Nash. 2020. “Variation in Outer Blubber Lipid Concentration Does Not Reflect Morphological Body Condition in Humpback Whales.” *Journal of Experimental Biology* 223(8). doi: 10.1242/jeb.213769.

Christiansen, Fredrik, Gísli A. Víkingsson, Marianne H. Rasmussen, and David Lusseau. 2013. “Minke Whales Maximise Energy Storage on Their Feeding Grounds.” *Journal of Experimental Biology* 216(3):427–36. doi: 10.1242/jeb.074518.

Currie, Jens J., Martin Van Aswegen, Stephanie H. Stack, Kristi L. West, and Fabien Vivier. 2021. “Rapid Weight Loss in Free Ranging Pygmy Killer Whales (Feresa Attenuata) and the Implications for Anthropogenic Disturbance of Odontocetes.” *Scientific Reports* 1–12. doi: 10.1038/s41598-021-87514-2.

Davis, Randall W. 2019. “Metabolism and Thermoregulation.” Pp. 57–87 in *Marine Mammals*.

Dawson, Stephen M., M. Hamish Bowman, Eva Leunissen, and Pascal Sirguey. 2017. “Inexpensive Aerial Photogrammetry for Studies of Whales and Large Marine Animals.” *Frontiers in Marine Science* 4(NOV):1–7. doi: 10.3389/fmars.2017.00366.

Dunkin, Robin C., William A. McLellan, James E. Blum, and D. Ann Pabst. 2005. “The Ontogenetic Changes in the Thermal Properties of Blubber from Atlantic Bottlenose Dolphin Tursiops Truncatus.” *Journal of Experimental Biology* 208(8):1469–80. doi:

CONFIDENTIAL

10.1242/jeb.01559.

Evans, Karen, Mark A. Hindell, and Deborah Thiele. 2003. “Body Fat and Condition in Sperm Whales, Physeter Macrocephalus, from Southern Australian Waters.” *Comparative Biochemistry and Physiology - A Molecular and Integrative Physiology* 134(4):847–62. doi: 10.1016/S1095-6433(03)00045-X.

Favilla, Arina B., Markus Horning, and Daniel P. Costa. 2022. “Advances in Thermal Physiology of Diving Marine Mammals: The Dual Role of Peripheral Perfusion.” *Temperature* 9(1):46–66. doi: 10.1080/23328940.2021.1988817.

Folkow, L. P., and A. S. Blix. 1992. “Metabolic Rates of Minke Whales (Balaenoptera Acutorostrata) in Cold Water.” *Acta Physiologica Scandinavica* 146(1):141–50. doi: 10.1111/j.1748-1716.1992.tb09402.x.

Fordyce, R. Ewan, and Felix G. Marx. 2018. “Gigantism Precedes Filter Feeding in Baleen Whale Evolution.” *Current Biology* 28(10):1670–76. doi: 10.1016/j.cub.2018.04.027.

Friedman, Matt, Kenshu Shimada, Larry D. Martin, Michael J. Everhart, Jeff Liston, Anthony Maltese, and Michael Triebold. 2010. “100-Million-Year Dynasty of Giant Planktivorous Bony Fishes in the Mesozoic Seas.” *Science* 327:990–94.

Galatius, Anders, and Carl Christian Kinze. 2016. “Lagenorhynchus Albirostris (Cetacea: Delphinidae).” *Mammalian Species* 48(933):35–47. doi: 10.1093/mspecies/sew003.

Gearty, William, Craig R. McClain, and Jonathan L. Payne. 2018. “Energetic Tradeoffs Control the Size Distribution of Aquatic Mammals.” *Proceedings of the National Academy of Sciences of the United States of America* 115(16):4194–99. doi: 10.1073/pnas.1712629115.

Glarou, Maria, Shane Gero, Alexandros Frantzis, José María Brotons, Fabien Vivier, Paraskevi Alexiadou, Margalida Cerdà, Enrico Pirotta, and Fredrik Christiansen. 2023. “Estimating Body Mass of Sperm Whales from Aerial Photographs.” *Marine Mammal Science* 39(1):251–73. doi: 10.1111/mms.12982.

Goldbogen, Jeremy A., Nicholas D. Pyenson, and Peter T. Madsen. 2023. “How Whales

CONFIDENTIAL

Dive, Feast, and Fast: The Ecophysiological Drivers and Limits of Foraging in the Evolution of Cetaceans." *Annu. Rev. Ecol. Evol. Syst. 2023* 54:307–25. doi: 10.1146/annurev-ecolsys-102220.

Gunnlaugsson, T., G. A. Víkingsson, S. D. Halldórsson, B. Þ. Elvarsson, T. Haug, and C. Lydersen. 2020. *Body Mass, Muscle, Blubber and Visceral Fat Content and Their Seasonal, Spatial and Temporal Variability in North Atlantic Common Minke Whales*. Vol. 21.

Hampton, I. F. G., G. C. Whittow, J. Szekerczes, and S. Rutherford. 1971. "Heat Transfer and Body Temperature in the Atlantic Bottlenose Dolphin, Tursiops Truncatus." *International Journal of Biometeorology* 15(2–4):247–53. doi: 10.1007/BF01803907.

Hampton, Ian F. G., and G. Causey Whittow. 1976. "Body Temperature and Heat Exchange in the Hawaiian Spinner Dolphin, Stenella Longirostris." *Comparative Biochemistry and Physiology Part A: Physiology* 55(2):195–97. doi: 10.1016/0300-9629(76)90092-X.

Heide-Jørgensen, Mads Peter, Lars Kleivane, Nils Øien, Kristin L. Laidre, and Mikkel Villum Jensen. 2001. "A New Technique for Deploying Satellite Transmitters on Baleen Whales: Tracking a Blue Whale (Balaenoptera Musculus) in the North Atlantic." *Marine Mammal Science* 17(4):949–54.

Hirtle, Nathan O., Julia E. F. Stepanuk, Eleanor I. Heywood, Fredrik Christiansen, and Lesley H. Thorne. 2022. "Integrating 3D Models with Morphometric Measurements to Improve Volumetric Estimates in Marine Mammals." *Methods in Ecology and Evolution* 13(11):2478–90. doi: 10.1111/2041-210X.13962.

Hokkanen, J. E. I. 1990. "Temperature Regulation of Marine Mammals." *Journal of Theoretical Biology* 145(4):465–85. doi: 10.1016/S0022-5193(05)80482-5.

Holyoake, Carly, Nahiid Stephens, and D. Coughran. 2012. "Collection of Baseline Data on Humpback Whale (Megaptera Novaeangliae) Health and Causes of Mortality for Long-Term Monitoring in Western Australia." in *48th Annual Conference of the Australian Marine Science Association*.

CONFIDENTIAL

IPCC. 2014. *Intergovernmental Panel on Climate Change, Fifth Assessment Report AR5*. Cambridge, UK: Cambridge University Press.

Irving, Laurence. 1973. “Aquatic Mammals.” Pp. 47–96 in *Comparative Physiology of Thermoregulation*. Academic Press.

Kanwisher, J., and G. Sundnes. 1965. “Physiology of a Small Cetacean.” *Hvalradets Skrifter* 48:45–53.

Kelley, Neil P., and Nicholas D. Pyenson. 2015. “Evolutionary Innovation and Ecology in Marine Tetrapods from the Triassic to the Anthropocene.” *Science* 348(6232). doi: 10.1126/science.aaa3716.

Kettemer, Lisa Elena, Audun H. Rikardsen, Martin Biuw, Fredrik Broms, Evert Mul, and Marie Anne Blanchet. 2022. “Round-Trip Migration and Energy Budget of a Breeding Female Humpback Whale in the Northeast Atlantic.” *PLoS ONE* 17(5 May). doi: 10.1371/journal.pone.0268355.

Kleiber, Max. 1932. “Body Size and Metabolism.” *Hilgardia* 6(11).

Koopman, H. N. 1998. “Topographical Distribution of the Blubber of Harbor Porpoises (Phocoena Phocoena).” *Journal of Mammalogy* 79(1):260–70. doi: 10.2307/1382862.

Kristensen, Henriette Gamborg. 2023. “Seasonal Observations, Site Fidelity and Movements of White Beaked Dolphins (Lagenorhynchus Albirostris) in Icelandic Coastal Waters.” Aalborg University.

Kshatriya, M., and R. W. Blake. 1988. “Theoretical Model of Migration Energetics in the Blue Whale, Balaenoptera Musculus.” *J. Theor. Biol* 133:479–98.

Kvadsheim, P. H., L. P. Folkow, and A. S. Blix. 1994. “A New Device for Measurement of the Thermal Conductivity of Fur and Blubber.” *Journal of Thermal Biology* 19(6):431–35. doi: 10.1016/0306-4565(94)90043-4.

Kvadsheim, P. H., L. P. Folkow, and A. S. Blix. 1996. “Thermal Conductivity of Minke Whale Blubber.” *Journal of Thermal Biology* 21(2):123–28. doi: 10.1016/0306-4565(95)00034-8.

CONFIDENTIAL

Lavigne, D. M., S. Innes, and Elizabeth F. Edwards. 1990. “Lower Critical Temperatures of Blue Whales, Balaenoptera Musculus.” *J. Theor. Biol* 144:249–57.

Lockyer, C. H., L. C. McConnell, and T. D. Waters. 1985. “Body Condition in Terms of Anatomical and Biochemical Assessment of Body Fat in North Atlantic Fin and Sei Whales.” *Canadian Journal of Zoology* 63(10):2328–38. doi: 10.1139/z85-345.

Lockyer, Christina, and Timothy Waters. 1986. “Weights and Anatomical Measurements of Northeastern Atlantic Fin (Balaenoptera Physalus, Linnaeus) and Sei (B. Borealis, Lesson) Whales.” *Marine Mammal Science* 2(3):169–85. doi: 10.1111/j.1748-7692.1986.tb00039.x.

Mackintosh, N., J. Wheeler, and A. Clowes. 1929. “Southern Blue and Fin Whales.” *Discovery Reports Vol. I* 257–540.

Magnúsdóttir, E. 2007. “Year Round Distribution and Abundance of White-Beaked Dolphins (Lagenorhynchus Albirostris) off the Southwest Coast of Iceland.” University of Iceland.

McGinnis, Samuel M., G. Causey Whittow, Carl A. Ohata, and Harriet Huber. 1972. “Body Heat Dissipation and Conservation in Two Species of Dolphins.” *Comparative Biochemistry and Physiology Part A: Physiology* 43(2):417–23. doi: 10.1016/0300-9629(72)90200-9.

McNab, B. 2012. *Extreme Measures: The Ecological Energetics of Birds and Mammals.* Chicago, IL: University of Chicago Press.

Meagher, Erin M., William A. McLellan, Andrew J. Westgate, Randall S. Wells, James E. Blum, and D. Ann Pabst. 2008. “Seasonal Patterns of Heat Loss in Wild Bottlenose Dolphins (Tursiops Truncatus).” *Journal of Comparative Physiology B: Biochemical, Systemic, and Environmental Physiology* 178(4):529–43. doi: 10.1007/s00360-007-0245-5.

Millar, J. S., and G. J. Hickling. 1990. “Fasting Endurance and the Evolution of Mammalian Body Size.” *Functional Ecology* 4(1):5. doi: 10.2307/2389646.

Miller, Carolyn A., Desray Reeb, Peter B. Best, Amy R. Knowlton, Moira W. Brown, and

CONFIDENTIAL

Michael J. Moore. 2011. “Blubber Thickness in Right Whales Eubalaena Glacialis and Eubalaena Australis Related with Reproduction, Life History Status and Prey Abundance.” *Marine Ecology Progress Series* 438:267–83. doi: 10.3354/meps09174.

Mitchell, Edward. 1973. *Draft Report on Humpback Whales Taken under Special Scientific Permit by Eastern Canadian Land Stations, 1969-1971*.

Moore, Jeffrey E., and Andrew J. Read. 2008. “A Bayesian Uncertainty Analysis of Cetacean Demography and Bycatch Mortality Using Age-at-Death Data.” *Ecological Applications* 18(8):1914–31. doi: 10.1890/07-0862.1.

Moore, Michael J., and Julie M. van der Hoop. 2012. “The Painful Side of Trap and Fixed Net Fisheries: Chronic Entanglement of Large Whales.” *Journal of Marine Biology* 2012:1–4. doi: 10.1155/2012/230653.

Moore, Michael, Amy Knowlton, Scott D. Kraus, William A. McLellan, and Robert K. Bonde. 2004. “Morphometry, Gross Morphology and Available Histopathology in North Atlantic Right Whale (Eubalaena Glacialis) Mortalities.” *Journal of Cetacean Research and Management* 6(3):199–214.

Morrison, Peter. 1962. “Body Temperatures in Some Australian Mammals. III. Cetacea (Megaptera).” *The Biological Bulletin* 123(1):154–69. doi: 10.2307/1539512.

Noren, Shawn R., and Randall S. Wells. 2009. *Blubber Deposition during Ontogeny in Free-Ranging Bottlenose Dolphins: Balancing Disparate Roles of Insulation and Locomotion*. Vol. 90.

Parry, D. A. 1949. “The Structure of Whale Blubber, and a Discussion of Its Thermal Properties.” *Journal of Cell Science* s3-90(9):13–25. doi: 10.1242/jcs.s3-90.9.13.

R Development Core Team. 2018. “R: A Language and Environment for Statistical Computing.”

Rasmussen, Marianne. 2009. “Whales in Skjálfandi Bay.” P. 16 in *Enviromental Impact Assessment (EIA) reports for Krafla Power Station, Bakki, Iceland*.

Robbins, Jooke, Luciano Dalla Rosa, Judith M. Allen, David K. Mattila, Eduardo R.

CONFIDENTIAL

Secchi, Ari S. Friedlaender, Peter T. Stevick, Douglas P. Nowacek, and Debbie Steel. 2011. “Return Movement of a Humpback Whale between the Antarctic Peninsula and American Samoa: A Seasonal Migration Record.” *Endangered Species Research* 13(2):117–21. doi: 10.3354/esr00328.

Rojano-Donãte, Laia, Birgitte I. McDonald, Danuta M. Wisniewska, Mark Johnson, Jonas Teilmann, Magnus Wahlberg, Jakob Højer-Kristensen, and Peter T. Madsen. 2018. “High Field Metabolic Rates of Wild Harbour Porpoises.” *Journal of Experimental Biology* 221(23). doi: 10.1242/jeb.185827.

Ryg, M., C. Lydersen, L. Ø. Knutsen, A. Bjørge, T. G. Smith, and N. A. Øritsland. 1993. “Scaling of Insulation in Seals and Whales.” *Journal of Zoology* 230:193–206.

Scholander, P. F., and William E. Schevill. 1955. “Counter-Current Vascular Heat Exchange in the Fins of Whales.” *Journal of Applied Physiology* 8(3):279–82. doi: 10.1152/jappl.1955.8.3.279.

Siebert, Ursula, Miguel L. Grilo, Tina Kesselring, Kristina Lehnert, Katrin Ronnenberg, Iwona Pawliczka, Anders Galatius, Line A. Kyhn, Michael Dähne, and Anita Gilles. 2022. “Variation of Blubber Thickness for Three Marine Mammal Species in the Southern Baltic Sea.” *Frontiers in Physiology* 13. doi: 10.3389/fphys.2022.880465.

Sigurðsson, Þorsteinn. 2005. “State of Marine Stocks in Icelandic Waters 2004/2005.” *Marine and Freshwater Research Institute* 1–189.

Stephens, Philip A., Ian L. Boyd, John M. McNamara, and Alasdair I. Houston. 2009. “Capital Breeding and Income Breeding: Their Meaning, Measurement, and Worth.” *Ecology* 90(8):2057–67. doi: 10.1890/08-1369.1.

Stevick, Peter T., Mariana C. Neves, Freddy Johansen, Marcia H. Engel, Allen Judith, Milton C. C. Marcondes, and Carole Carlson. 2011. “A Quarter of a World Away: Female Humpback Whale Moves 10 000 Km between Breeding Areas.” *Biology Letters* 7(2):299–302. doi: 10.1098/rsbl.2010.0717.

Strunk, Thomas H. 1971. *Heat Loss from a Newtonian Animal*. Vol. 33.

Struntz, D. J., W. A. McLellan, R. M. Dillaman, J. E. Blum, J. R. Kucklick, and D. A.

CONFIDENTIAL

Pabst. 2004. “Blubber Development in Bottlenose Dolphins (Tursiops Truncatus).” *Journal of Morphology* 259(1):7–20. doi: 10.1002/jmor.10154.

Sumich, James L. 1986. “Latitudinal Distribution, Calf Growth and Metabolism, and Reproductive Energetics of Gray Whales, Eschrichtius Robustus.” Oregon State University.

Sumich, James L. 2021. “Why Baja? A Bioenergetic Model for Comparing Metabolic Rates and Thermoregulatory Costs of Gray Whale Calves (Eschrichtius Robustus).” *Marine Mammal Science* 37(3):870–87. doi: 10.1111/mms.12778.

Tang, Bin, Yujiang Hao, Chaoqun Wang, Zhengyu Deng, Guilin Shu, Kexiong Wang, and Ding Wang. 2023. “Variation of Blubber Thickness of the Yangtze Finless Porpoise (Neophocaena Asiaeorientalis Asiaeorientalis) in Human Care: Adaptation to Environmental Temperature.” *Water Biology and Security*. doi: 10.1016/j.watbs.2023.100200.

Víkingsson, Gísli A., and Mads Peter Heide-Jørgensen. 2015. “First Indications of Autumn Migration Routes and Destination of Common Minke Whales Tracked by Satellite in the North Atlantic during 2001-2011.” *Marine Mammal Science* 31(1):376–85. doi: 10.1111/mms.12144.

Vongraven, Dag, Morten Ekker, Arild R. Espelien, and Frode J. Aarvik. 1990. “ Postmortem Body Temperatures in the Minke Whale, Balaenoptera Acutorostrata .” *Canadian Journal of Zoology* 68(1):140–43. doi: 10.1139/z90-020.

Watts, Peter, Stephen Hansen, and D. M. Lavigne. 1993. “Models of Heat Loss by Marine Mammals: Thermoregulation Below the Zone of Irrelevance.” *Journal of Theoretical Biology* 163(4):505–25. doi: 10.1006/jtbi.1993.1135.

Wells, RandallS., HowardL. Rhinehart, LarryJ. Hansen, JayC. Sweeney, ForrestI. Townsend, Rae Stone, David R. Casper, MichaelD. Scott, AletaA. Hohn, and TeriK. Rowles. 2004. “Bottlenose Dolphins as Marine Ecosystem Sentinels: Developing a Health Monitoring System.” *EcoHealth* 1(3). doi: 10.1007/s10393-004-0094-6.

Wisniewska, Danuta Maria M., Mark Johnson, Jonas Teilmann, Laia Rojano-Doñate, Jeanne Shearer, Signe Sveegaard, Lee A. A. Miller, Ursula Siebert, and Peter

CONFIDENTIAL

Teglberg T. Madsen. 2016. “Ultra-High Foraging Rates of Harbor Porpoises Make Them Vulnerable to Anthropogenic Disturbance.” *Current Biology* 26(11):1441–46. doi: 10.1016/j.cub.2016.03.069.

Woodward, Becky L., Jeremy P. Winn, and Frank E. Fish. 2006. “Morphological Specializations of Baleen Whales Associated with Hydrodynamic Performance and Ecological Niche.” *Journal of Morphology* 267(11):1284–94. doi: 10.1002/jmor.10474.

Worthy, G. A. J., and E. F. Edwards. 1990. “Morphometric and Biochemical Factors Affecting Heat Loss in a Small Temperate Cetacean (Phocoena Phocoena) and a Small Tropical Cetacean (Stenella Attenuata).” *Physiological Zoology* 63(2):432–42. doi: 10.1086/physzool.63.2.30158506.

Yasui, W. Y., and D. E. Gaskin. 1986. “Energy Budget of a Small Cetacean, the Harbour Porpoise, Phocoena Phocoena (L.).” *Ophelia* 25(3):183–97. doi: 10.1080/00785326.1986.10429749.

Zhang, Changqun, Haojie Zhou, Fredrik Christiansen, Yujiang Hao, Kexiong Wang, Zhangbing Kou, Ruipeng Chen, Jie Min, Randall Davis, and Ding Wang. 2023. “Marine Mammal Morphometrics: 3D Modeling and Estimation Validation.” *Frontiers in Marine Science* 10. doi: 10.3389/fmars.2023.1105629.

CONFIDENTIAL

Figure Legends

Figure 1. (A) Relative body widths (expressed as % of BL) for all species, at each measurement site (% BL from the rostrum) across the body. The solid lines represent the mean widths for each measurement site of each species (see colour key). (B) Relative body heights (expressed as % of BL) for all species, at each measurement site (% BL from the rostrum) across the body. The solid lines represent the mean heights for each measurement site of each species (see colour key). (C) Height-to-width ratio at each measurement site (% BL from the rostrum) across the body of all five species. The ratio of 1:1 (circular cross-sectional body shape) is indicated by the dashed red line. The solid lines represent the mean HW ratios of each species (see colour key). Sample size of measured cetaceans: harbour porpoises: N = 50, white-beaked dolphins: N = 294, minke whales: N = 30, humpback whales: N = 282, blue whales: N =29.

Figure 2. Comparative plot showing the relationship between surface-area-to-volume ratio (SVR) and total body length across all target species. The different colours represent the different species (see colour key), while the dotted red line represents a SVR of 1. The solid black lines represent the back-transformed fitted values of the linear models for each species.

Figure 3. Boxplots displaying (A) the total body length and (B) the predicted blubber thickness (relative to BL) distribution among five cetacean species. Pp = harbour porpoises (N = 50); La = white-beaked dolphins (N = 294); Ba = minke whales (N = 30); Mn = humpback whales (N = 282); Bm = blue whales (N = 29).

Figure 4. (A) Absolute heat loss and (B) volume-specific heat loss, as a function of total body length for the five target species (see colour key). The solid black lines represent the back-transformed fitted values of the linear models for each species.

Figure 5. Partial effects plots showing the effect of surface area, body temperature, thermal conductivity of blubber, and blubber thickness on absolute heat loss rates (Fig. A, C, E, G) and on volume-specific heat loss rates (relative) among the five target species (Fig. B, D, F, H). The different colours represent the different species examined in this study (see colour key). The solid black lines represent the back-transformed fitted values of the linear models for each species.

Figure 6. (A) Absolute heat loss rates, and (B) volume-specific heat loss rates for all target species (see colour key), as a function of body length, using (i) blubber thickness predictions based on the mean relative blubber thickness measured at the flank, at the midpoint of their lateral side below the dorsal fin (M4 point) (solid lines) and (ii) blubber thickness predictions based on the upper and lower 95% CI relative blubber thickness values at the M4 point for each species (dotted lines).

CONFIDENTIAL

Figure 1

Click here to access/download;Figure;Fig1_MorphometricsPlot_means.png

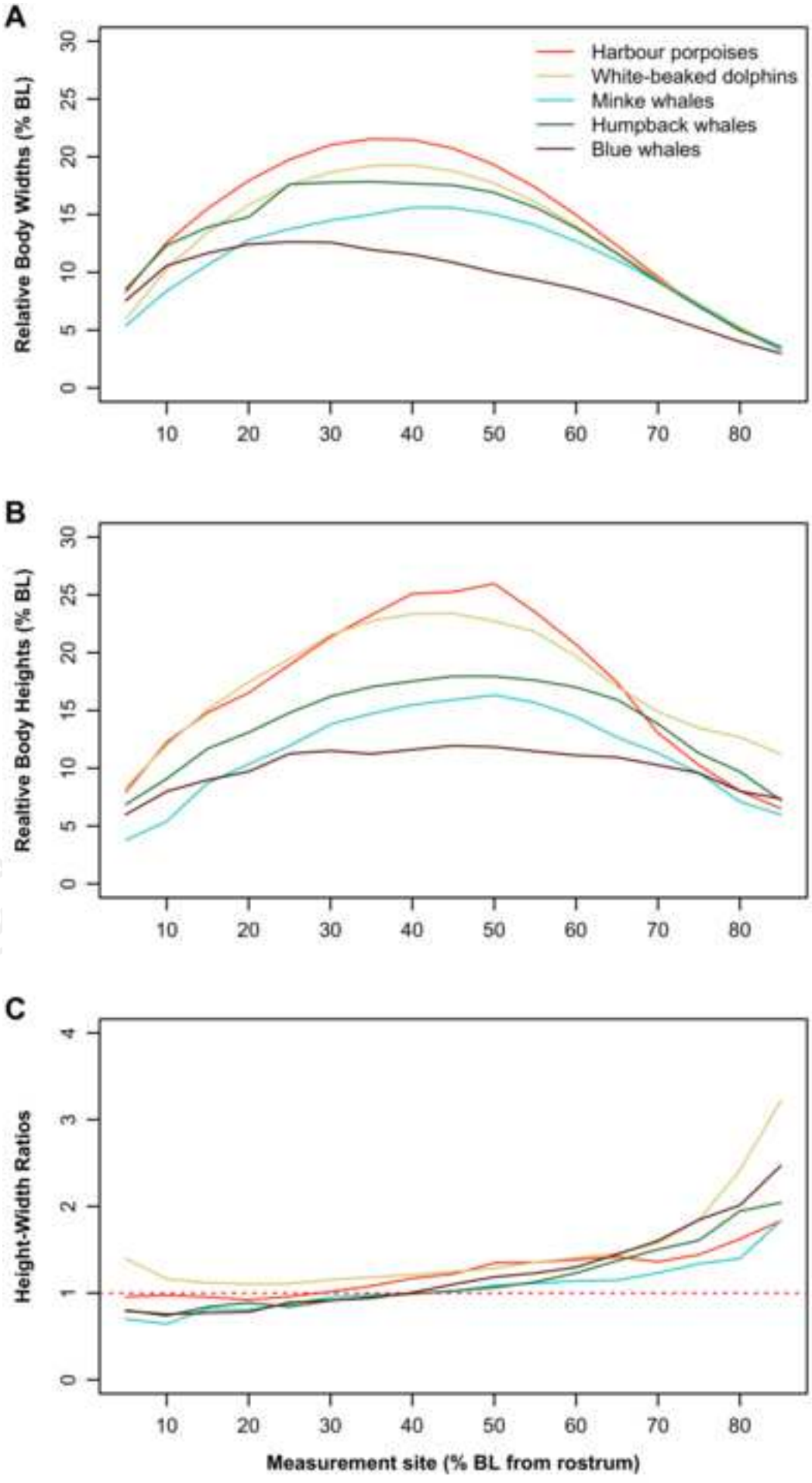


Figure 2

Click here to access/download;Figure;Fig2_SVratios_all.png

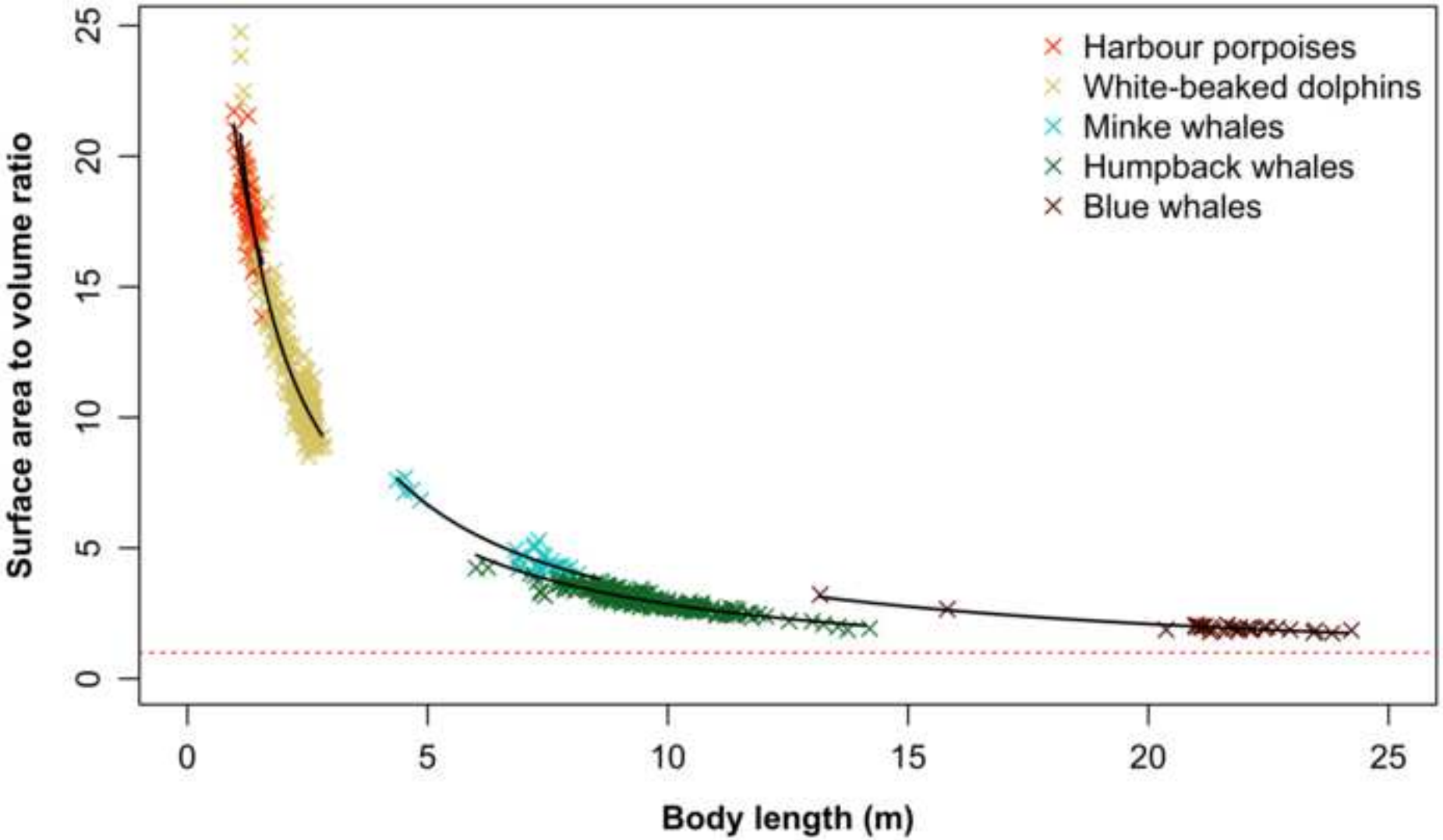


Figure 3

Click here to access/download;Figure;Fig3_Boxplots_BL_BT.png

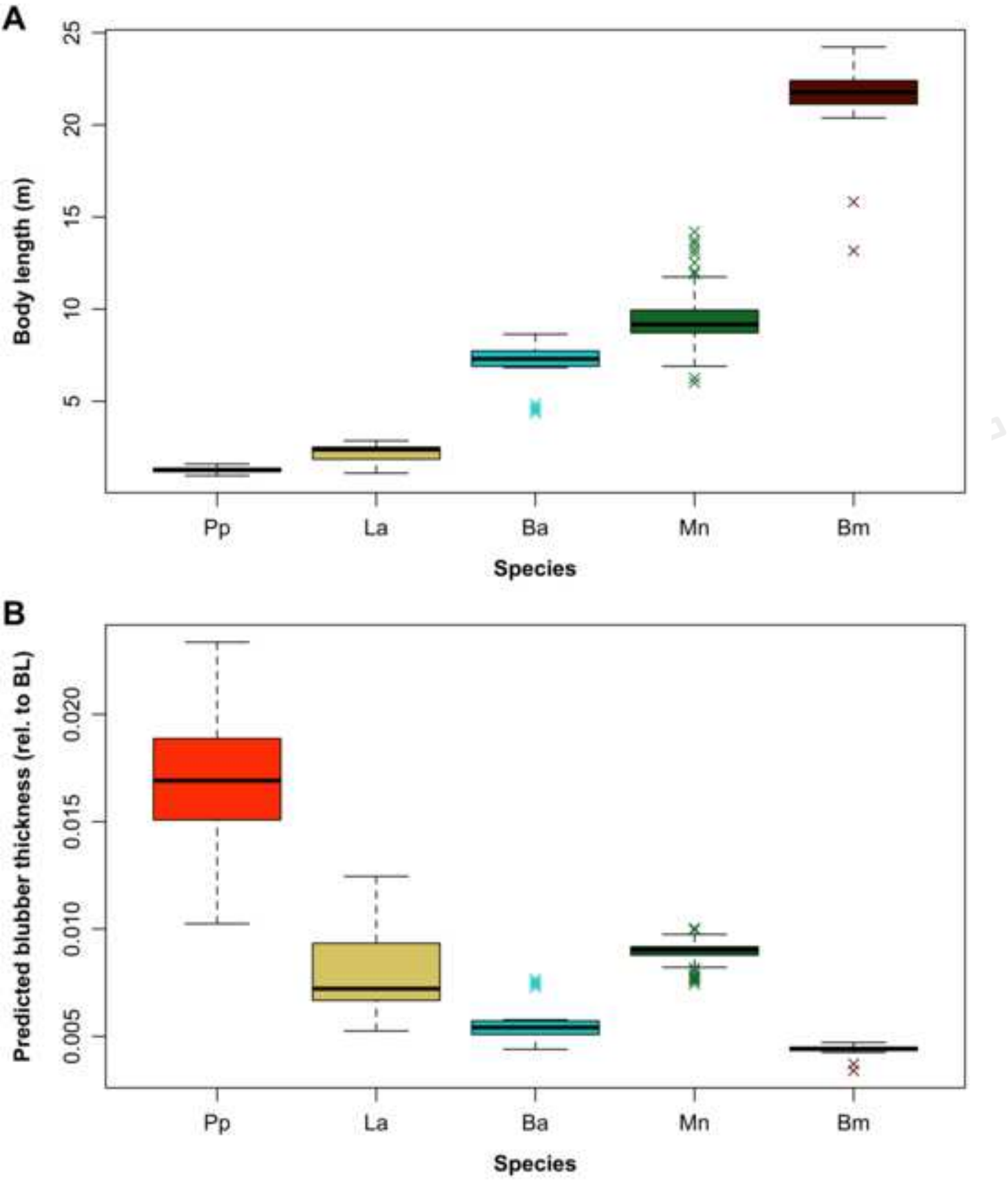


Figure 4

Click here to access/download;Figure;Fig4_HeatLoss_relM4.png

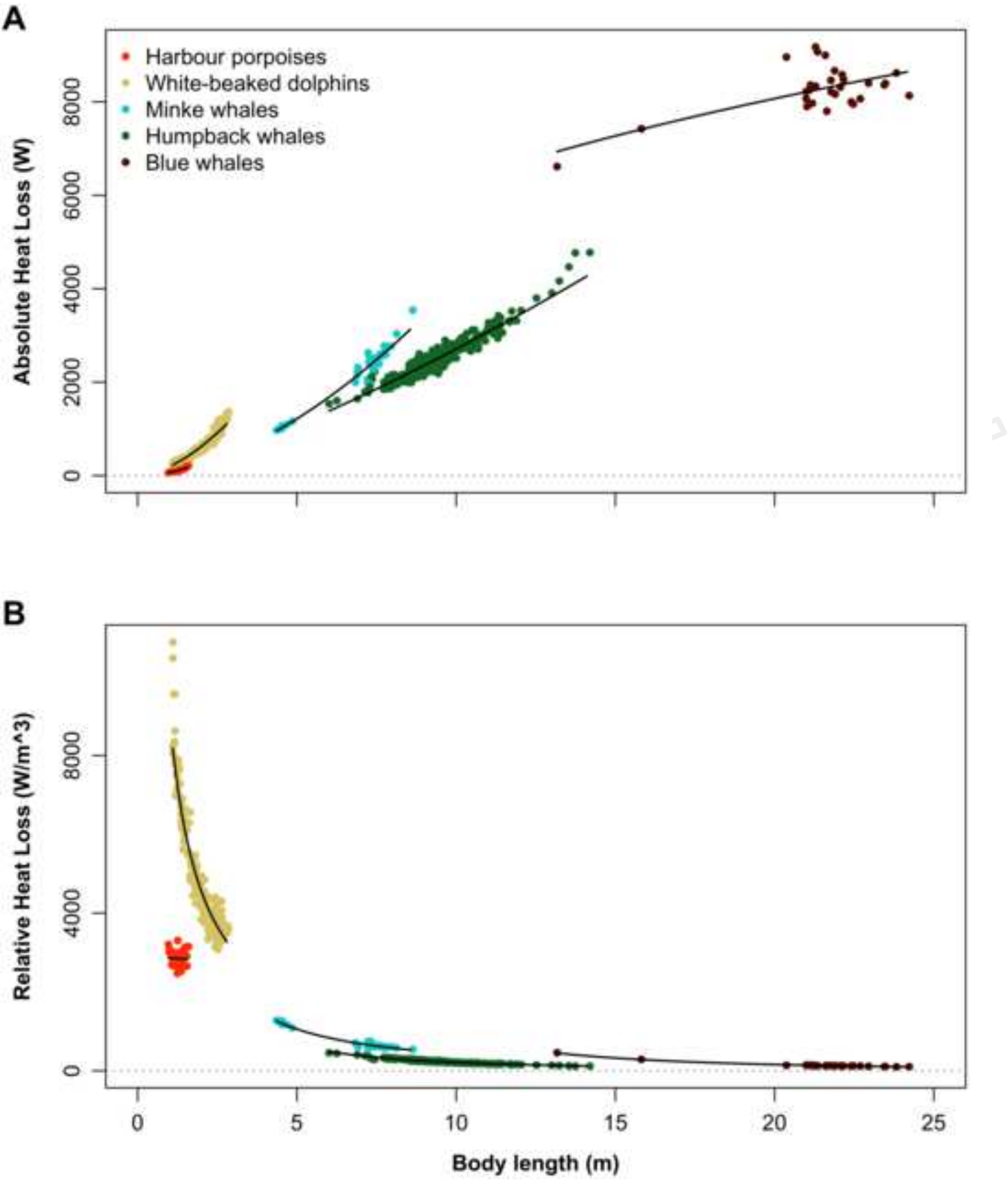


Figure 5

Click here to access/download;Figure;Fig5_HeatLoss_PartialEffectsPlot.png

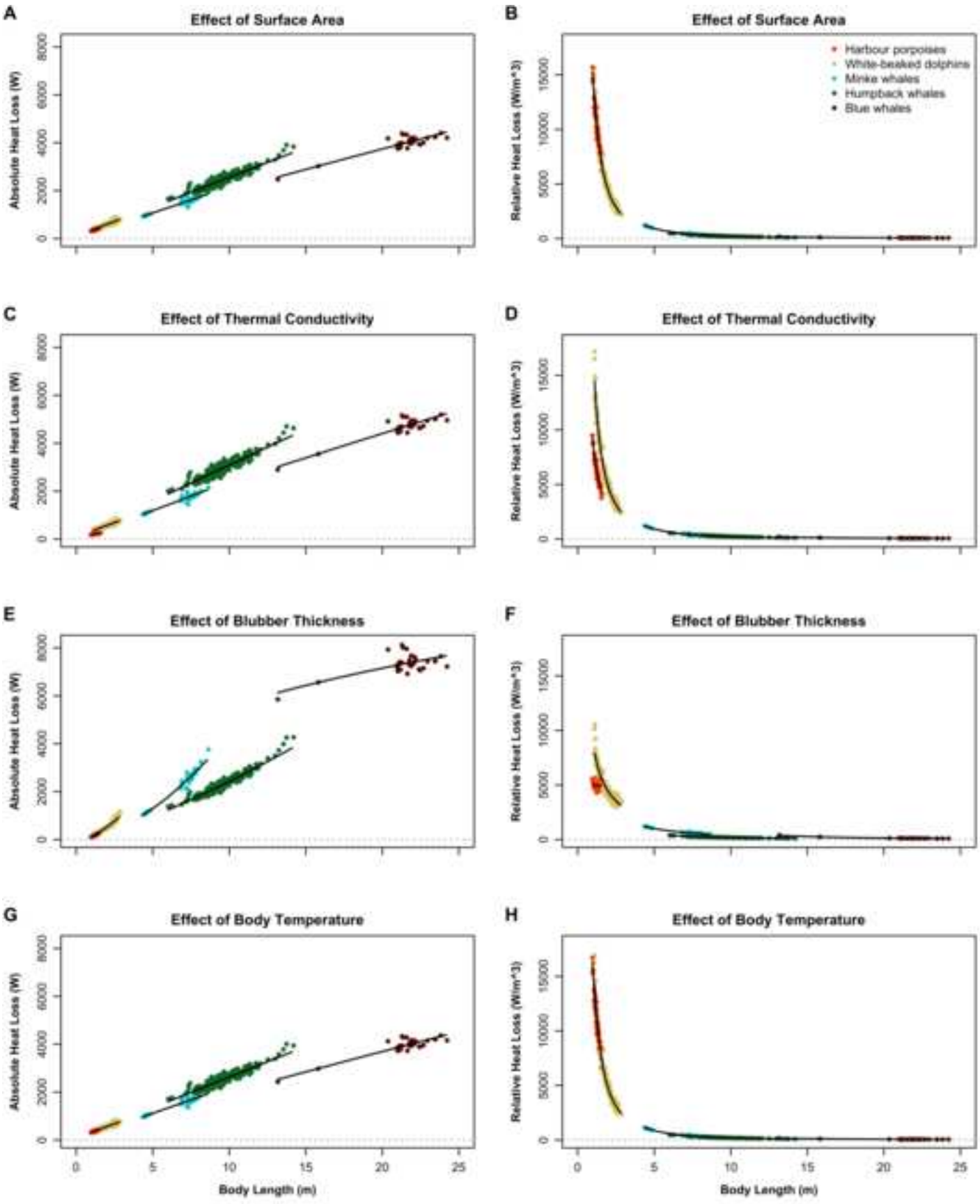


Figure 6

Click here to access/download;Figure;Fig6_HeatLoss_M4extremes.png

